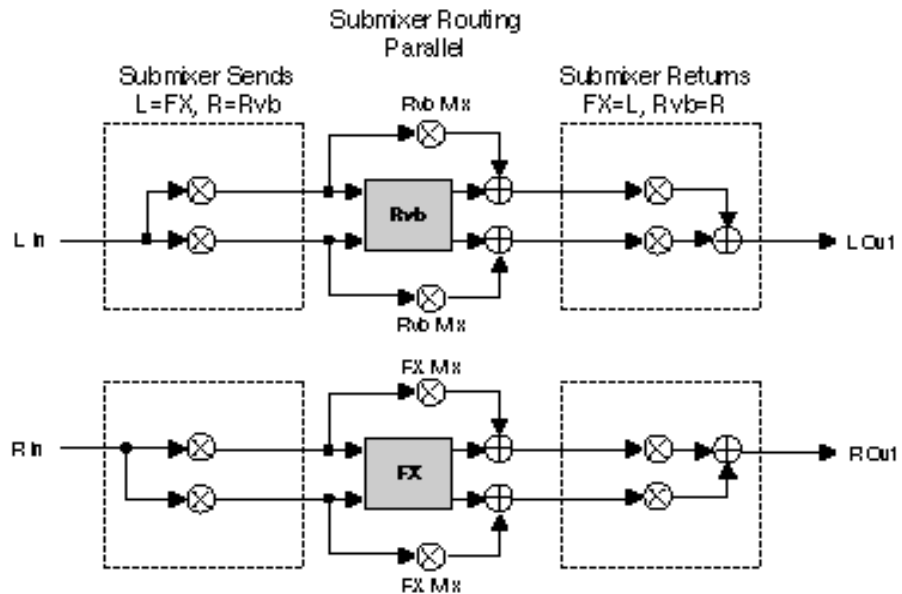


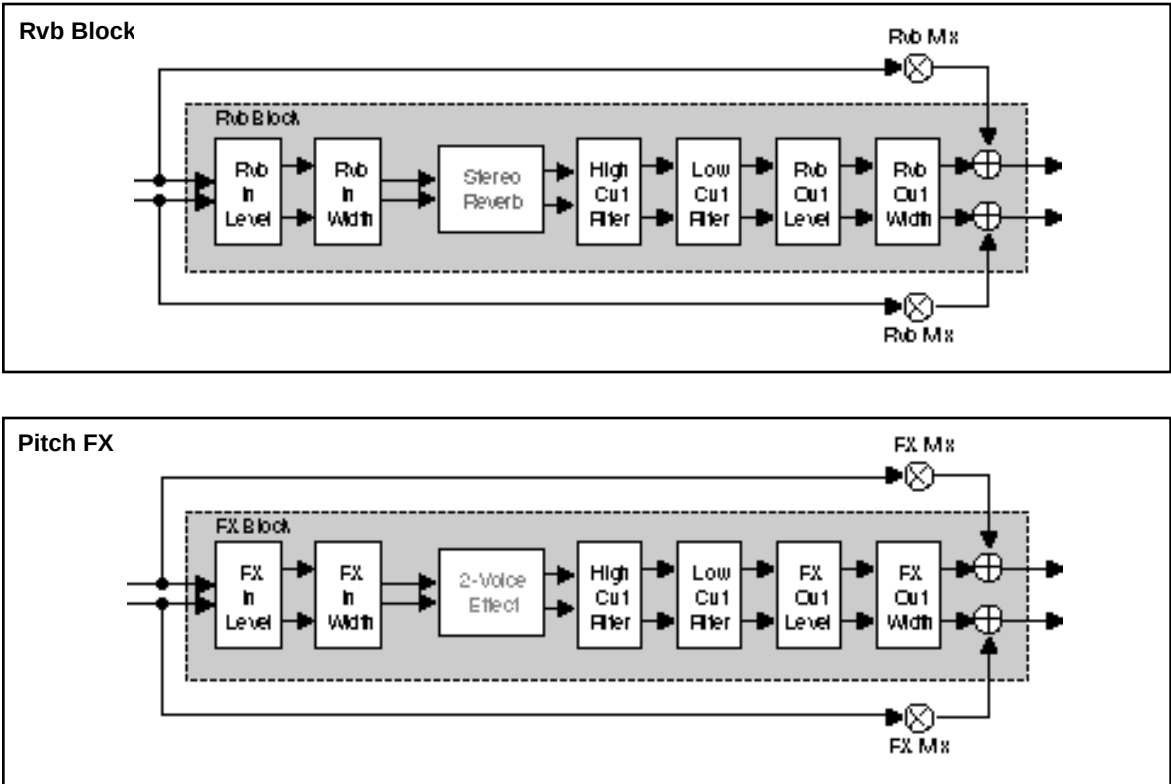
**Dual Mono In/Mono Out**

- The Rvb block processes only the left input as a mono signal, and the FX block processes only the right input as a mono signal.
- A mono mix of the Rvb block output is sent to the left main output. A mono mix of the FX output is sent to the right main output.

**Submixer Sends****1.0 100 L=Rvb, R=FX****Submixer Returns****1.1 100 Rvb=L, FX=R****Submixer Routing****1.2 0 Parallel**

The Rvb and FX blocks shown in the previous configurations represent the Stereo Reverb and 2-Voice Pitch Shift effects. Each of these blocks has its own set of controls, shown in the diagrams below.

**Rvb and FX Block Controls**



These effect block controls are also found in the Submixer row of each parameter matrix.

|          |        |         |         |     |       |       |       |       |        |        |
|----------|--------|---------|---------|-----|-------|-------|-------|-------|--------|--------|
| 1        | 1.0    | 1.1     | 1.2     | 1.3 | 1.4   | 1.5   | 1.6   | 1.7   | 1.8    | 1.9    |
| Submixer | 6 ends | Returns | Routing | Mix | InLvl | In Wd | HiCut | LoCut | OutLvl | Out Wd |

These parameters provide independent control of input and output level and width, as well as high cut and low cut parameters for each FX and Rvb block.

Press **Load/Q** to toggle between Rvb and FX parameters.

These parameters are described below, in the order in which they appear in the Submixer row. (All other effect parameters are identical to those described earlier for the 4-Voice and 6-Voice algorithms.

**RvbMix, FX Mix**

These are independent wet/dry mix controls for the Rvb and FX effect blocks.

**RvbInLvl, FX InLvl**

These are independent input level controls for the Rvb and FX blocks.

**Rvb In W, FX In W**

These are input width controls for the Rvb and FX blocks. These are essentially the same as the PCM 81 FX Width and Rvb Width parameters. The difference is that the width control in the Pitch FX algorithms is located at the input to the effect. This makes it possible to independently alter the stereo image of material feeding the two effect blocks.

**RvbHiCut, FX HiCut**

These parameters provide independent 6dB per octave, stereo high-cut filters on the outputs of the Rvb and FX blocks.

**RvbLoCut, FX LoCut**

These parameters provide independent 6dB per octave, stereo low-cut filters on the outputs of the Rvb and FX blocks.

**RvbOutLvl, FX OutLvl**

These are independent output level controls for the Rvb and FX blocks.

**Rvb Out W, FX Out W**

These parameters allow independent control of the stereo image of the output of each effect block.

The Vocal Fix Pitch Correct algorithm is designed to work with monophonic (one note at a time) vocal sources. The algorithm contains an intelligent pitch shifter combined with a PCM 81 Chamber reverb. The intelligent pitch shifter detects the pitch of incoming audio and produces effects based on the detected pitch.

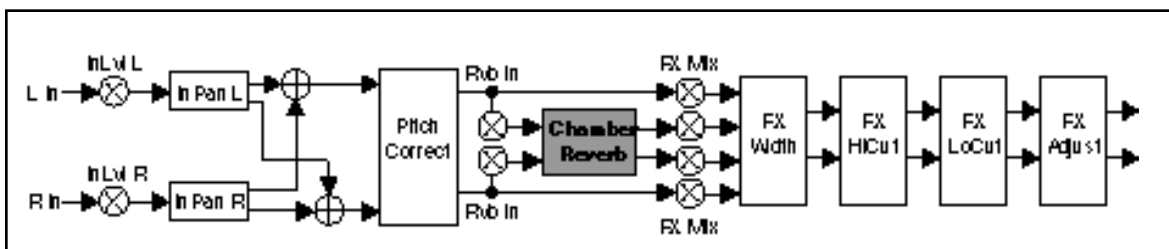
### The Pitch Correct Algorithm and Its Parameters

The reverb follows the pitch shifter in series. The FX Mix parameter is set to 0% reverb as most applications require only pitch processing.

The algorithm provides a simple, yet powerful, set of tools for correction of previously recorded melody tracks which are off pitch. The Pitch Detect display shows the pitch of the input source in real time, displayed as a chromatic note and  $\pm$ cents so that you can see exactly where and how much correction is needed. For most applications correction will only be needed for notes that are audibly sour.

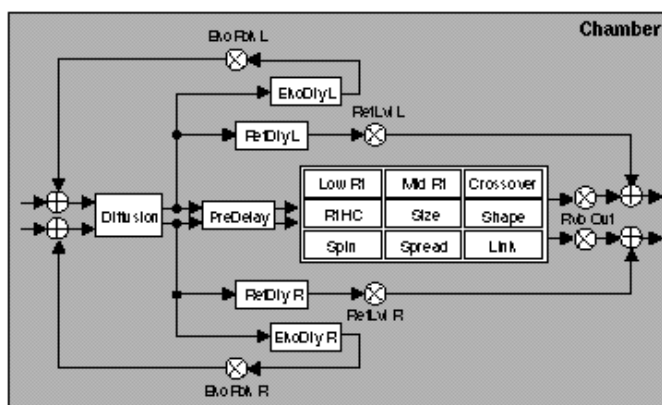
There are two basic approaches to this type of pitch correction. The first is to patch the Correct parameter to a switch or to the ADJUST knob, the second is to use a MIDI keyboard to control correction. The patch method is preferable for fine tuning a performance, whereas using a MIDI keyboard provides more flexibility when handling more serious pitch errors, or when it is necessary to completely rework the melody.

MIDI control is straightforward. Just connect a MIDI keyboard to the PCM 81 MIDI IN jack, set the keyboard and the PCM 81 to the same MIDI channel, and you're ready to start. Don't try to play the melody note-for-note — just press the keys needed to correct bad notes. When you press a key on the MIDI keyboard, the audio is forced to the keyboard's pitch. When you release the key, the audio is processed according to the settings you have made to Correct and Tracking. Typically, Correct will be set to 0% and Tracking set to Fastest. This results in the MIDI note determining the correct pitch and triggering correction on and off. (By the way, you don't need a score to know which note, just watch the Pitch Detect display.)



|                 |             |                |               |              |                  |                |                 |           |         |         |
|-----------------|-------------|----------------|---------------|--------------|------------------|----------------|-----------------|-----------|---------|---------|
| 0 Controls      | 0.0 Mix     | 0.1 LoAdj      | 0.2 rnlwLIR   | 0.3 FX Mix   | 0.4 rFX HILo Cut | 0.5 FX Width   | 0.6 rLo HILimit |           |         |         |
| 1 Reverb Time   | 1.0 LowR    | 1.1 Mid R      | 1.2 Crossover | 1.3 R HC     | 1.4 Pre Delay    | 1.5 rRel LowL  | 1.6 rBw RMLD    |           |         |         |
| 2 Reverb Design | 2.0 Size    | 2.1 Diffusion  | 2.2 Shape     | 2.3 Spread   | 2.4 Spin         | 2.5 Link       | 2.6 rReb Out Th |           |         |         |
| 3 Pitch         | 3.0 Detect  | 3.1 Correct    | 3.2 GDRsp     | 3.3 Tracking | 3.4 Low Pitch    | 3.5 High Pitch | 3.6 Tuning      | 3.7 6pIke |         |         |
| 4 GHI           | 4.0 Conts   | 4.1 6emones    |               |              |                  |                |                 |           |         |         |
| 5 Modulation    | 5.0 rModLFO | 5.1 rModRR Env | 5.2 Mod EnvL  | 5.3 Mod EnvR | 5.4 rModLch      | 5.5 rModSml    | 5.6 rModSmlC    |           |         |         |
| Patches         | Patch 0     | Patch 1        | Patch 2       | Patch 3      | Patch 4          | Patch 5        | Patch 6         | Patch 7   | Patch 8 | Patch 9 |

The Chamber reverb produces an even, relatively dimensionless reverberation, with little change in color as the sound decays. The initial diffusion is similar to the Concert Hall algorithm, but the sense of space and size is much less obvious. This characteristic, along with the low color in the decay tail, makes Chamber useful on a wide range of material. It is especially useful on spoken voice, giving a noticeable increase in loudness with very low coloration.



**The Pitch Correct  
Parameters**

**Controls Row**

**FX HiCut, FX LoCut**

These parameters control stereo high cut and low cut filters on the algorithm outputs. Each has a range of 20Hz- 20kHz.

**Detect**

This parameter displays the source used for pitch detection: Input, Fixed or MIDI.

**Pitch Row**

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input | Pitch is detected from the sum of the left and right input signals. The detected pitch and the number of cents sharp or flat is displayed and updated in real time.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Fixed | ADJUST scrolls through the range of available pitches. The number of cents sharp or flat from the selected fixed pitch is displayed and updated in real time.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| MIDI  | This setting is displayed when you use a MIDI keyboard as the note detect source. Connect a MIDI keyboard to PCM 81 rear panel MIDI IN jack and set both to the same MIDI channel. Whenever a preset that uses the PitchCorrect algorithm is running, the shifter will detect the pitch of any MIDI key you press. The display will show the MIDI pitch and the number of cents sharp or flat in real time as the pitch of the input signal varies. Correct the pitch of any melody by simply playing the appropriate MIDI key when ever the melody pitch drifts from the desired note. |

**Correction**

This parameter lets you control the amount of off-pitch correction (0-100%) applied. When set to 0, no correction is applied. When set to 100%, notes are shifted as close as possible to the frequency of the detected pitch (or to the frequency of harmonies of the detected pitch).

**GldResp**

This parameter is similar to the GldResp parameter found in the Master Delay parameter of several PCM 81 delay and reverb algorithms. It sets the pitch shifter's responsiveness to the pitch detect source and also to modulation by internal controllers (LFO, AR etc.) and MIDI controllers (aftertouch, velocity, etc.) At the highest setting (100), the pitch will track the detect source very closely. At lower settings, the pitch will lag behind the modulation source. A setting of about 50 generally yields good results for tracking vocals.

**Tracking**

This parameter controls how quickly (accurately) the pitch detector will follow pitch changes in the detect source. This control can be set to Fastest, Fast, Moderate, Slow or Hold.

|                         |                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Fastest                 | This is generally the most useful setting as it provides accurate tracking as well as good handling of vibrato or pitch bends. |
| Fast, Moderate and Slow | These settings are useful for special effects.                                                                                 |
| Hold                    | This setting freezes at the last detected pitch, effectively turning any melody into a pedal tone.                             |

### Low Pitch and High Pitch

These controls allow you to set a range of pitches to be detected. This is very useful with source material that has extraneous noise, for example, a vocal track which contains some headphone leakage picked up by the vocal mike. If Low and High Pitch are set to bracket only the vocal frequencies, there will be no glitches caused by the shifter detecting headphone material outside the vocal range.

Note that Low Pitch also affects the amount of time that the pitch shifted material is delayed. At the lowest setting, the amount of delay is about 60ms. At the highest setting it is about 20ms.

For reference, a list of the lowest and highest pitches of some typical sources is shown below.

|                 |            |
|-----------------|------------|
| vocal: bass     | Eb2 - -Eb4 |
| vocal: baritone | Ab2 - -Ab4 |
| vocal: tenor    | C3 - -C5   |
| vocal: alto     | F3 - -F5   |
| vocal: soprano  | C4 - -C6   |

### Tuning

This parameter allows you to set a reference frequency for the pitch, "A". The range is from 410.0-470.0Hz. The standard reference frequency is 440.0Hz.

### Splice

This parameter allows you to change the amount of time used to crossfade the splices that occur whenever pitch is shifted. A value of 8ms is appropriate for most material. Higher values will produce longer/smoothier cross-fades, but the intelligibility of transients and high frequency content may be affected. Lower values will reproduce transients more accurately and have more high frequency content, but the splices may be more audible.

## Shift Row Shift Cents

This parameter determines the amount of fixed shift for the stereo input channels. The value is given in cents (1 cent=1/100th of a semitone, 1200 cents/octave). The range is adjustable in increments of a single cent. This value can be modulated smoothly with patch sources to create vibrato and other pitch-related special effects.

### Shift Semitones

This parameter provides  $\pm 1$  octave of pitch shift in half-step (chromatic) intervals.

Note: As the Cents and Semitones controls are additive, it is possible to shift tones by as much as  $\pm 2$  octaves.

## The Presets

The PCM 81 has 300 factory-designed presets which are organized into six banks of 50 each (labeled **P0-P5**). Each bank is organized in a matrix of 5 rows of 10. Press the front panel **Program Banks** button to display the first bank. Press it again to switch to another bank. Simultaneously press **Program Banks** and either the **Up** or **Down** button to backstep through the banks. The display will show the bank label and the matrix location, the preset name, and the algorithm from which the effect is derived. Turn SELECT to scroll through all of the presets in numerical order. Use **Up** or **Down** to jump forward or backward by 10. Press **Load/★** to load any displayed preset.

In the Program Banks mode (as in the Register Banks mode), ADJUST is a soft knob. Each preset has one or more parameters patched to this knob, providing a quick way to make useful changes to the effect. (A ■ symbol in the upper left corner of the display indicates that the currently running effect has an ADJUST knob patch.) When you turn ADJUST, the display will show the name assigned to ADJUST, as well as the patch value. Continue turning ADJUST to alter the patch value along its available range. The program banks are organized as follows:

| Bank      | Matrix Location | Preset Type                     |
|-----------|-----------------|---------------------------------|
| <b>P0</b> | 0.0–1.9         | Multi Effects                   |
|           | 2.0–3.5         | Modulation Effects              |
|           | 3.6–4.9         | Special Effects                 |
| <b>P1</b> | 0.0–3.6         | Rhythmic Echo and Delay Effects |
|           | 3.7–4.9         | Ambience Effects                |
| <b>P2</b> | 0.0–1.0         | EQ Effects                      |
|           | 1.1–2.4         | Spatial Effects                 |
|           | 2.5–4.0         | Gain Effects                    |
|           | 4.1–4.9         | Resonant Chord Effects          |
| <b>P3</b> | 0.0–2.8         | Reverb Effects                  |
|           | 2.9–3.9         | Processed Reverb Effects        |
|           | 4.0–4.9         | Remix Effects                   |
| <b>P4</b> | 0.0–1.0         | Vocal Shift                     |
|           | 1.1–1.9         | Vocal Harmony                   |
|           | 2.0–2.9         | Guitar Magic                    |
|           | 3.0–3.9         | Instrument Shift & Harmony      |
|           | 4.0–4.9         | Percussion & Tempo              |
| <b>P5</b> | 0.0–0.9         | Sci-Fi FX                       |
|           | 1.0–1.9         | Pitch & Delay                   |
|           | 2.0–2.9         | Pitch Sequences                 |
|           | 3.0–3.7         | Pads & Drones                   |
|           | 3.8–3.9         | Utility Programs                |
|           | 4.0–4.9         | Pitch Correct                   |

Each preset is described in this section with a header which indicates the matrix location, the program name, the name assigned to the ADJUST knob, and the range of ADJUST knob control. This header is followed by a brief description of the effect. For additional reference, at the end of this chapter is an alphabetical list of the presets, with references to bank and matrix location.



## Program Bank 0 (P0)

### Multi Effects (P0 0.0 – P0 1.9)

- 0.0 Prime Blue** **ADJUST: Efx/Rvb X 0–127**  
A combination of 3 stereo effects: 6 voice chorus, rhythmic echoes, and reverb. You can dial-in the exact proportion of each. As ADJUST is turned from 0 to 127 the effect smoothly changes from chorus only, to chorus with echoes, to chorus with echoes and reverb, to reverb with echoes, and finally to reverb only. Use **Tap** to set the echo rhythms.
- 0.1 EkoChorus** **ADJUST: FX Mix 0–127**  
A 6 voice stereo chorus with echoes and stereo reverb. ADJUST controls the mix of chorus/echoes with reverb. The Soft Row provides all the tweaks you'll need to get the most out of this classic multi effect.
- 0.2 Wet Chorus** **ADJUST: FX Mix 0–100**  
This is a thick wet chorus combined with a small amount of reverb to produce a large lush sound. It can fatten up any track and is especially good for adding body to guitar. ADJUST controls the combination of delays and chorus with reverb.
- 0.3 Wet Chorus 2** **ADJUST: FX Mix 0–100**  
This is a variation of Wet Chorus with a subtle amount of delays and reverb to produce just the right amount of ambience. ADJUST controls the combination of delays and chorus with reverb.
- 0.4 Wet Chorus 3** **ADJUST: FX Mix 0–100**  
A variation of Wet Chorus 2 with even thicker chorus (bordering on a slight flange) and slightly modified delays. ADJUST controls the combination of delays and chorus with reverb.
- 0.5 Chorus Plate** **ADJUST: KorusMix 0–100**  
Combination of smooth rolling delays in the reverb shell combined with a more aggressively modulated reverb tail. ADJUST changes the non-modulated to modulated reverb tail. Twisted for guitar.
- 0.6 Chorus Room** **ADJUST: Decay 0–80**  
Slow chorus applied to one side of the reverb. The speed of the modulation follows input level. ADJUST controls the reverb decay.
- 0.7 ChorusToRvb** **ADJUST: Decay 0–127**  
Input level controls the relative mix of stereo chorus and reverb. Loud signals are mostly chorus and the reverb level comes up as the signal fades away. ADJUST controls the reverb decay time.
- 0.8 Funkus Room** **ADJUST: LFO Rate 0–100**  
This preset works great on electronic pianos / keyboards. Strong modulation applied to both delays and reverb tail. ADJUST controls the speed of the LFO modulation.
- 0.9 Detune & Room** **ADJUST: Detune 0–127**  
A very tight room combined with separate pitch modulation on the left and right inputs. ADJUST controls the depth of the detuning.
- 1.0 Detune&Dbl** **ADJUST: Detune 0–127**  
Similar to Detune & Room, but several delay voices have been added to produce doubling effects. ADJUST controls the amount of detune on each side. Soft Row goodies include a master delay control that allows you to vary the time difference between the delay voices.
- 1.1 Tight Space** **ADJUST: SplitWide 0–100**  
A bright, dry ambience combined with left and right detune and delay. ADJUST controls the amount of detune as well as the left/ right delay spread.

- 1.2 Flange >Rvb** **ADJUST: FX Mix 0–100** **Multi Effects (P0 0.0 – P0 1.9) cont'd.**  
 This stereo effect feeds the output of a flanger into a concert hall reverb. ADJUST controls the mix of dry and wet flanged audio. The Soft Row includes master delay and feedback parameters for adding echoes, as well as parameters for modulation and image control.
- 1.3 Flange+Rvb** **ADJUST: FX Mix 0–100**  
 A rich 6 voice chorus in parallel with reverb. ADJUST controls the mix of the two independent stereo effects. The Soft Row includes the essential parameters for creating many variations of this classic combination of effects.
- 1.4 X Eko Flange** **ADJUST: X-Fbk 0–100**  
 A stereo flanger with rhythmic echoes feeding into stereo reverb. ADJUST controls the cross-feedback of the echoes (which causes them to bounce from side to side as they repeat). Press **Tap** to synchronize the echoes with rhythmic sources.
- 1.5 Glide > Verb** **ADJUST: FX Mix 0–100**  
 Three stereo effects in series: gliding delays, rhythmic echoes and reverb. ADJUST controls the mix of delay effects and reverb. Essential controls for each effect are included in the Soft Row.
- 1.6 Glide X-Ekos** **ADJUST: Eko Time 0–100**  
 Similar to Glide > Verb, but ADJUST lets you dial in echo times from 0 to 2 seconds. Feedback and cross feedback are combined to create echo patterns that change as they repeat.
- 1.7 Detuned Ekos** **ADJUST: Ekos/Beat 1–24**  
 Plate reverb combined with independent stereo delays. The delay voices are diffused and modulated. The echoes are rhythmic — set the tempo with **Tap** or MIDI clocks. Use ADJUST to set the number of echoes per beat.
- 1.8 StereoEqEkos** **ADJUST: Ekos/Beat 1–24**  
 Six rhythmic echo voices are EQ'd and panned across stereo space. ADJUST sets the number of repeats per beat.
- 1.9 ADJUpMyEchos** **ADJUST: EchosLvl 0–10**  
 Concert Hall reverb with 4 panned echo voices in the background. ADJUST controls the level of the echo voices. Press **Tap** to synchronize the echoes with rhythmic material.
- 2.0 FSw2 Elevate** **ADJUST: Feedback 0–99** **Modulation Effects (P0 2.0 – P0 3.5)**  
 Similar to Detuned Ekos, multiple echo voices with diffusion, modulation and a touch of plate reverb. In this effect, Foot Switch 2 is patched to turn on the AR Envelope which, in turn, sweeps the master delay. ADJUST controls the feedback of these rhythmic echoes.
- 2.1 Chorus &Pan** **ADJUST: LFO Rate 0–100**  
 This moving chorus shifts the output signal from left to right, back and forth. ADJUST sets the speed. Increasing the Delay Master parameter will smear the delay images.
- 2.2 Chorus&Amb** **ADJUST: MstChorus 0–100**  
 Similar to Chorus&Pan but with an added ambience. ADJUST controls the speed and depth of the chorus.
- 2.3 6 Vox Chorus** **ADJUST: Less/More 0–50**  
 Starting point for all chorus sounds. ADJUST controls both the chorus depth and speed. Use the Delay Master to open or close the spacing between the 6 delay voices.

**Modulation Effects  
(P0 2.0 – P0 3.5) cont'd.****2.4 Split C&E ADJUST: InputPan 0–100**

The left input is processed into a lush 3-voice chorus with the voice panners adjusted from center to left. The right input is processed into a rhythmic 3-voice echo with the output panned from center to right. ADJUST cross-pans the inputs. 0 = left/right stereo, 50 = mono, 100 = right/left stereo.

**2.5 Env:PanKorus ADJUST: Chorus 0–127**

The AR Envelope drives the speed of the pan based on the presence or absence of input signal. ADJUST varies the chorus amount.

**2.6 6 Vox Flange ADJUST: MstrDepth 0–100**

A rich stereo flanger with a touch of reverb. ADJUST controls the flange depth.

**2.7 Mod Max ADJUST: Mod Knob 0–50**

Multiple parameters all being modulated together. With a stereo input the LFO drives the input pans to each delay and reverb processor. A definite "twist your head off" effect. ADJUST controls several aspects of the modulation.

**2.8 PreciseGlide ADJUST: Resonance 0–100**

A very clean stereo gliding delay and reverb. ADJUST controls the resonance of the glide. Soft Row includes master delay and feedback parameters so you can add stereo echoes to the effect.

**2.9 Round 147 ADJUST: RotorRate 0–20**

Big wooden rotary speaker cabinet miked fairly close. ADJUST sets the speed of the rotors. FX Mix is available in the soft row to control the amount of reverb relative to the total effect output.

**3.0 FSw2 Rotary ADJUST: Width 0–100**

A dual-rotor speaker cabinet with a very wide stereo spin and a touch of ambience. Use ADJUST to control the width and direction of the spinning rotors. 0= very wide left-to-right, 64 = mono, 127 = very wide right-to-left. Foot Switch 2 is patched to the Latch to toggle between slow and fast speeds. The AR envelope is used to simulate the inertial drag as the rotors speed up or slow down.

**3.1 RotorCabinet ADJUST: Slow/Fast 0–1**

Similar to FSw2 Rotor, but ADJUST is used to toggle the speed of the rotors between slow and fast. Soft Row parameters include FX Width which sets the width and direction of the spin.

**3.2 MIDI Rotary ADJUST: EkoFbk 0–100**

Similar to FSw2 Rotor, but with echoes added and the rotors patched up for MIDI control. MIDI After Touch toggles speed. Press hard to spin fast, press hard again to spin slow. ADJUST controls the amount of echo feedback. Use **Tap** to synchronize the echoes with rhythmic material.

**3.3 Tiled Rotary ADJUST: Slow/Fast 0–1**

The RotorCabinet effect in a tiled room. Try it with background vocals, as well as the usual keyboard and guitar sources. ADJUST toggles the spin rates between slow and fast.

**3.4 RotoWood ADJUST: Speed 0–10**

ADJUST quickly advances the speed of the rotors. Increase the Feedback Master for more upper rotor "howling".

**3.5 RandomImages ADJUST: ImageKnob 0–127**

This effect works best with solo instruments or voices. Individual input notes come out at random locations in the stereo image. ADJUST varies delay time, adds chorusing, and turns up the level of tempo related echoes. Delays "creep" slowly out to new time values.

- 3.6 Under Water** **ADJUST: DrownKnob 0–100** **Special Effects (P0 3.6– P0 4.9)**  
This effect really pulls you under! It will submerge any track under water. ADJUST controls the over all rate of the effect.
- 3.7 Thunder FX** **ADJUST: FlashTime 0–127**  
An unusual special effect that produces a rolling clap of thunder from a percussive source (tom toms, etc.) and ethereal sweeps from synth pads. The effect is driven by the AR Env, which is available in the soft row. ADJUST controls the AR release rate.
- 3.8 Thunder FX 2** **ADJUST: FlashTime 0–127**  
A variation of Thunder FX. The zaps are produced by setting a much faster release rate. ADJUST controls the release rate.
- 3.9 ChaosImpuls** **ADJUST: GldResp 0–100**  
This special effect is hard to describe but interesting to listen to. It produces chaotic rhythms and detunings from the input source. ADJUST sets the chaotic limits. Try this with individual percussive hits.
- 4.0 DemonDescent** **ADJUST: GldResp 0–100**  
Dark reverb and modulated detuning are combined to create an eerie special effect. ADJUST controls the rate and depth of the detuning.
- 4.1 RoughIdle FX** **ADJUST: Speed 0–127**  
Use this to mutate any steady, broad band source into a poorly tuned combustion engine. ADJUST controls the speed.
- 4.2 RoxStutter** **ADJUST: FX Mix 0–100**  
Use on those strong rock and roll lead vocals. The trailing echoes on voices 3 and 4 go longer as the signal disappears. ADJUST combines the delay effects with a plate reverb effect. Delays 2 and 5 are available in the soft row to add more echo presence.
- 4.3 Split Pitch?** **ADJUST: SplitWide 0–127**  
Modulated varispeed in stereo. One side goes low as the other side goes high. ADJUST controls the rate and depth of the varispeed. Good for special FX processing of various sources including synths and dialog.
- 4.4 TryTalk'n** **ADJUST: GldResp 0–100**  
A variation of Split Pitch? In this version highly colored reverb and additional delay voices have been added. ADJUST controls the rate and depth of the varispeed.
- 4.5 AutoInfinite** **ADJUST: FX Mix 0–100**  
With signal present the reverb time runs long. With signal absent the infinite process is switched on. The threshold for the event is set high such that after a source is running in the infinite process, you can play softer passages against it which chorus and echo at the same time. AR Envelope parameters are included in the soft row. Adjust the threshold to suit your application.
- 4.6 Remove Cntr** **ADJUST: Cntr Fc 0–127**  
This effect uses filters and crosstalk cancellation to remove mono material from a stereo mix. ADJUST controls a low pass filter for the center channel frequencies. As it is turned from 0-127 mono material is added back into the mix.
- 4.7 V-Eliminate** **ADJUST: Cut Zone 0–127**  
This effect is a vocal eliminator. It is similar to Remove Cntr, in that it will remove mono material from a stereo mix. In this effect, however, only frequencies in the vocal range are removed. This keeps mono low and high frequencies (kick, bass, snare cymbals, etc.) in the mix. ADJUST controls the width of the elimination band. Note that phase cancellation will occur if the two outputs are summed to mono.

### Special Effects (P0 3.6– P0 4.9) cont'd.

- 4.8 NoCenter Eko** **ADJUST: Center Fc 0–127**  
This is the same effect as Remove Center with delays added to the processed signal. It allows you to add echoes to the left and right material without affecting the mono material of a stereo mix or sub mix. For example, you can add additional echoes to a mix without affecting the vocals, kick or snare. ADJUST controls a low pass filter for the center channel frequencies.
- 4.9 Rvb On L–R** **ADJUST: Decay 0–127**  
Similar to NoCenterEko, this effect adds dense plate reverb to the left and right material without affecting the mono material in a stereo mix. ADJUST controls the mid Rt of the reverberator.

## Program Bank 1 (P1)

### Rhythmic Echo and Delay Effects (P1 0.0 – P1 3.6)

NOTE: The echoes in all of the Rhythmic Echo and Delay effects can be synchronized to tempo. Tempo is set by dialing it in, pressing **Tap**, or selecting MIDI Clock as the Tempo. (Tempo mode 0.2) Be sure to try these effects synchronized with MIDI sequence and drum patterns.

- 0.0 [n]Ekos/Beat** **ADJUST: [n] 1–24**  
This preset produces stereo echoes with a touch of reverb. ADJUST controls the number of echoes per beat. Soft Row goodies include controls for master scaling of the delay rhythms and feedback, EQ and panning, as well as reverb parameters.
- 0.1 StereoTapDly** **ADJUST: Mstr Fbk 0–100**  
A variation of [n]Ekos/Beat this effect has left and right delay voices. ADJUST controls master feedback for both voices. The Soft Row includes controls for rhythm, feedback and panning for each voice
- 0.2 OffBeat Eko** **ADJUST: DarkKnob 0–127**  
This variation of [n]Ekos/Beat produces a syncopated echo rhythm pattern. ADJUST controls high cut filters in the echo feedback paths – great for “tape” echo effects.
- 0.3 OffBeat Eko2** **ADJUST: DarkKnob 0–127**  
Additional rhythmic delay voices are used to create a more complicated syncopation pattern than OffBeat Eko. ADJUST controls high cut filters in the echo feedback paths – great for “tape” echo effects.
- 0.4 Dotted 8ths** **ADJUST: EchoSlope 0–127**  
Rhythmic delay voices produce a dotted 8th-note pattern. ADJUST controls the slope of the first six repeats. 0 = loud to soft (normal echoes), 64 = six repeats of equal loudness, 127 = soft to loud (reversed echoes).
- 0.5 Shuffle Ekos** **ADJUST: Shuffl Lvl 0–127**  
This effect creates echoes with a shuffle feel. One repeat on the beat followed by one repeat off the beat. ADJUST controls the relative levels of the on-beat and off-beat echoes. 0 = on-beat echo only, 127 = off-beat echo only.
- 0.6 Shuffles** **ADJUST: EchoSlope 0–127**  
Rhythmic delay voices produce a shuffle pattern. ADJUST controls the slope of the first six repeats. 0 = loud to soft (normal echoes), 64 = six repeats of equal loudness, 127 = soft to loud (reversed echoes).

**0.7 Sliding Eko****ADJUST: Slide It! 0–100**

This stereo delay effect lets you dial in the perfect “feel” to match the moment. Two echoes are produced. One is fixed on the beat. The other can be slid in musical time anywhere in front of or behind the beat by turning ADJUST. 0-49 = in front of the beat, 50 = on the beat, 51–100 = behind the beat. Of course, the Soft Row contains additional controls for fine tuning the effect.

**Rhythmic Echo and Delay Effects****(P1 0.0 – P1 3.6) cont'd.****0.8 Six Trips****ADJUST: EchoSlope 0–127**

Rhythmic delay voices produce a six note triplet pattern. Each repeat gets progressively louder. ADJUST controls the slope of the first six repeats. 0 = loud to soft (normal echoes), 64 = six repeats of equal loudness, 127 = soft to loud (reversed echoes).

**0.9 6 StrokeRoll****ADJUST: Attack 0–127**

Turn a single drum hit into a six-stroke roll with this effect. ADJUST controls the attack of the first five repeats. 0 = loud to soft (normal echoes), 64 = five repeats of equal loudness, 127 = soft to loud (reversed echoes).

**1.0 TapGated Rvb****ADJUST: Gate dB 0–85**

This effect uses a tempo-controlled LFO as a rhythmic gate to the inputs of a large stereo chamber. The gate is opened every other beat for a duration of one eighth-note. Use this to add reverb to selected beats of a stereo source. (Try this one synchronized to MIDI with a drum machine as a source.) ADJUST controls the input levels in dB to the reverb when the gate is closed. 0 = always open (0dB), 85 = fully closed (-85dB). The Soft Row includes the LFO parameters so you can adjust the overall timing of the gate.

**1.1 TapEkoGate 1****ADJUST: GateWidth 0–127**

This effect uses a tempo-controlled LFO as a rhythmic gate to the inputs of a delay and reverb combination. The gate is opened once every four beats for a duration of one eighth-note. ADJUST determines the portion of the four beats in which the gate is open. 0 = open for all four beats, 127 = open for only the first 16th note of four beats. The Soft Row includes the LFO parameters so you can adjust the overall timing of the gate.

**1.2 TapEkoGate 2****ADJUST: GateWidth 0–127**

This is a variation of TapEkoGate 1 with different echo rhythms. ADJUST determines the portion of the four beats in which the gate is open. 0 = open four all four beats, 127 = open for only the first 16th note of four beats. The Soft Row includes the LFO parameters so you can adjust the overall timing of the gate.

**1.3 TapEkoClear****ADJUST: GateWidth 0–127**

This preset is similar to the two TapEkoGate presets, except that the rhythmic gate is also patched to clear the delay voices when it opens. This allows very high delay feedback values to be used without creating long echo trails. ADJUST determines the portion of the four beats in which the gate is open. 0 = open for all four beats, 127 = open for only the first 16th note of four beats. The Soft Row includes the LFO parameters so you can adjust the overall timing of the gate.

**1.4 Tap Chamber1****ADJUST: Width 0–90**

This preset is a tempo controlled variation of SnareChamber. The effect is a tight chamber with liveness that is constantly changing. The change is tempo controlled. Press **Tap** to synchronize it to the beat. ADJUST controls the width of the reverb within the overall stereo image of the effect (0 = mono, 45 = stereo, 90 = surround). Use this to open up sampled drum mixes.

**1.5 Tap Chamber2****ADJUST: Decay 0–127**

This is a version of Vox Chamber with a rhythmic twist. The width of the reverb, within the overall stereo image of the effect, is changed rhythmically by Sw 1. This change is tempo controlled. Press **Tap** twice to synchronize to the beat. ADJUST controls reverb decay.

**Rhythmic Echo and  
Delay Effects  
(P1 0.0 – P1 3.6) cont'd.**

- 1.6 LatchedEkos** **ADJUST: EchoWidth 1–99**  
The inputs and outputs of stereo delays are gated on and off by two rhythmic switches. The Latch is used to trigger the AR envelope, which in turn alternates the left and right delay outputs. ADJUST controls how long the inputs to the delays remain open over a period of four beats.
- 1.7 X-PanEQ BPM** **ADJUST: Low Tone 0-60**  
This effect cross-panns the inputs between independent left and right band pass filters and adds some bright, highly diffused reverb. The pan rate is tempo controlled. ADJUST controls the low frequency limits of both bands.
- 1.8 Pan->Eko BPM** **ADJUST: DarkKnob 0–127**  
This effect pans the inputs across the stereo inputs of a reverb and independent left and right delays, each with slightly different EQ. The pan rate is tempo controlled. ADJUST controls the high frequency limits of both bands.
- 1.9 Tempo Verb** **ADJUST: Liveness 0–40**  
In this tempo-based reverb effect, decay changes in tempo. Slow tempos yield longer reverb time; fast tempos result in shorter decay. Set tempo with two presses of **Tap**, or with MIDI.
- 2.0 Tempo Gate** **ADJUST: High Cut 0-50**  
This heavily gated reverb effect has duration set by tempo. **Tap** in two quarter-notes to set the gate duration.
- 2.1 Tape Echo** **ADJUST: DarkKnob 0–127**  
This preset simulates a stereo tape echo. The echo rate is tempo controlled. Press **Tap** to synchronize with rhythmic material. ADJUST controls high frequency damping – the echoes get darker as they repeat. The Soft Row includes parameters for adding and reverb as well additional controls for the delays and filters.
- 2.2 NonLinear 1** **ADJUST: EchoSlope 0–127**  
This effect produces echoes with non linear decay. ADJUST controls the decay slope. 0 = loud to soft (normal echoes), 64 = six repeats of equal loudness, 127 = soft to loud (reversed echoes). In this preset the slope is set to produce a simple reversed echo effect. The repeat rate is tempo controlled. The Soft Row contains parameters for controlling the stereo image, adding reverb, scaling the repeat rate, and more.
- 2.3 NonLinear 2** **ADJUST: EchoSlope 0–127**  
Similar to NonLinear 1, with some additional processing. Chorus and reverb have been added, and the delays bounce from side to side as they repeat. In this preset the slope is set to 64 to produce repeats of equal loudness. The Soft Row includes parameters for controlling the stereo image, chorus depth and rate, etc.
- 2.4 NonLinear 3** **ADJUST: EchoSlope 0–127**  
This is a variation of NonLinear 2 with more radical processing and an inverted slope.
- 2.5 Six Across** **ADJUST: Spacing 0-100**  
This multi-tap delay effect filters each voice to a specific bandpass with each tap positioned successively across the panoramic spectrum. ADJUST controls the overall spacing between the voices.
- 2.6 BandEko Rvb** **ADJUST: Center Fc 0–127**  
In this preset, the outputs of a stereo band pass filter are sent to left and right rhythmic delays. These delays are in the reverb diffusor loop. The effect produces diffuse, highly colored echoes and reverb. Use ADJUST to set the center frequency of the band pass filter.

**2.7 BandEkoSweep****ADJUST: Mstr Fbk 0–100**

A variation of BankEko Rvb. In this preset, the center frequency of the band pass filter is swept by the LFO producing echoes of shifting colors. Reverb and diffusion are turned off, but you can add them in from the Soft Row. ADJUST controls master feedback for the left and right delays.

**Rhythmic Echo and****Delay Effects****(P1 0.0 – P1 3.6) cont'd.****2.8 LFO EQ Echo****ADJUST: LC Depth 0–100**

A stereo rhythmic effect created by modulating low cut and high cut filters with the LFO sine and cosine waves. Interesting on rhythm tracks and individual instruments. ADJUST controls the depth of the low cut modulation.

**2.9 Chase Echo****ADJUST: ChaseRate 0–100**

This is LFO EQ Echo with panning added to the filtered delay outputs. ADJUST controls the pan rate.

**3.0 Panned Dlys****ADJUST: Mstr Fbk 0–100**

In this effect, a pair of delays are panned to produce echoes that drift across stereo space. ADJUST controls the feedback of both delays. More goodies await you in the Soft Row.

**3.1 X-Pan Delays****ADJUST: Mstr Fbk 0–100**

The outputs of the delay pair are cross-panned in this effect. Mono material will produce echoes that repeat, but remain in the center. Stereo material will slowly swap left and right as it repeats. Use ADJUST to set the feedback for both delays.

**3.2 Dly>EQ>Pans****ADJUST: Mstr Fbk 0–100**

The input signal is split into two frequency bands by this effect. The two bands are then sent through two different rhythmic delays whose outputs are panned in opposite directions. This effect deconstructs the tone of input material, except when the panning and delay rhythms align in the center of the stereo field. Try it with a drum mix or other broad band material. Use ADJUST to set the feedback for both delay voices.

**3.3 Ekoz 4 Drums****ADJUST: FX Mix 0–100**

This preset is a space setter for percussive instruments. It combines a short plate reverb effect with four high density echoes which are highly diffused. ADJUST varies the amount of delay effect to reverb effect.

**3.4 Haas PanKnob****ADJUST: L=0, R=127 0–127**

A panner that uses the slight differences in left and right channel delay to produce panning without changing relative levels. Use ADJUST to pan from left to right.

**3.5 Dial a Delay****ADJUST: Delay 0–100**

This one is your basic delay. Use ADJUST to dial-in up to 2.5 seconds of stereo delay. The Soft Row will take you beyond the basics, with parameters for reverb design, EQ and feedback.

**3.6 PrecisionDly****ADJUST: L ms/100 0–100**

Use this preset if you need precise alignment of left and right channel audio. ADJUST allows you to offset the left channel by up to 1 ms in 100 increments. The Soft Row contains two sets of additional delay controls for the left and right channels, each with 1ms resolution, as well as independent level and panning controls.



**Ambience Effects**  
**(P1 3.7 – P1 4.9)**

- 3.7 PhoneOrRoom?** **ADJUST: Pick One 0–1**  
 Use ADJUST to choose between a mono telephone filter and a small room with stereo ambience. The Soft Row provides access to the filter controls as well as reverb design parameters.
- 3.8 CheapTV Room** **ADJUST: The Walls 1–10**  
 This stereo preset simulates the sound of a Lo-Fi TV in a small room. Use ADJUST to change the reflectivity of the walls. The Soft Row provides access to the filter controls as well as reverb design parameters.
- 3.9 Empty Stage** **ADJUST: Liveness 0–100**  
 Made for creating live sounding spaces. ADJUST opens the space to be more reflective and airy.
- 4.0 Tomb Room** **ADJUST: The Walls 1–10**  
 Use this ambience preset to place source material within a very reflective tomb. ADJUST moves the source deeper into this scary space.
- 4.1 Comb Room** **ADJUST: Tone 0–100**  
 This effect provides a tunable comb filter and reverb to produce a highly colored ambience. Use ADJUST to change the tuning of the comb filter.
- 4.2 Zoom Over** **ADJUST: Speed 1–25**  
 Run an effect (or even a continuous synthesizer drone) into this preset, and the sound will approach you from the center, spread out to the sides as it passes overhead, and recede into the distance behind you. ADJUST controls the speed of the fly-by. This effect will image properly in either 2-channel or Surround mixes.
- 4.3 OneShotCarBy** **ADJUST: Speed 1–100**  
 Input level triggers this left-to-right drive-by. When signal is detected, it will move from left to right. Doppler pitch shift is simulated as the image moves across the stereo field. Use ADJUST to set the speed of the drive-by. Once audio has been detected, the effect will cycle once. To drive by again, stop and restart the audio track. To reverse the direction of the effect, set the Width parameter to –45 — you'll find it in the Soft Row.
- 4.4 AmbientCarBy** **ADJUST: Speed 1–100**  
 This is a variation of OneShotCarBy. In this preset the effect is enhanced with the addition of some stereo ambience. Also, in this variation the effect repeats automatically. ADJUST controls the speed of the drive-by.
- 4.5 HeadOn CarBy** **ADJUST: Speed 1–100**  
 This preset uses less ambience than ZoomOver, and is more appropriate for things that approach on the road than in the air. ADJUST sets the speed of the effect. This effect will image properly in either 2-channel or Surround mixes.
- 4.6 2WayStreet** **ADJUST: Speed 1–100**  
 Another variation of AmbientCarBy. This preset produces both left-to-right and right-to-left drive-by effects — like listening to the sound of traffic on a two-way street. The speed for each direction is slightly different. ADJUST is a master speed control for both eastbound and westbound traffic.
- 4.7 IntoTunnel** **ADJUST: Speed 1–25**  
 This preset simulates the sound of a source approaching you from the side, passing you and then entering a tunnel. ADJUST controls the speed of the source. To reverse the direction of the effect, set the Width parameter to –45 — you'll find it in the Soft Row.

**4.8 2WayTunnel****ADJUST: Speed 1–100**

This is a variation of IntoTunnel. The source approaches and enters the tunnel, then turns around and comes back. Use ADJUST to control the speed.

**Ambience Effects****(P1 3.7 – P1 4.9) cont'd.****4.9 FinishLine****ADJUST: Speed 1–100**

This preset adds two pairs of stereo delays to the basic drive-by effect to simulate the 1st, 2nd, and 3rd place cars crossing the finish line. ADJUST controls the speed. Be sure to try this and the other drive-by effects with different types of source material. Almost any continuous source will produce interesting spatial sound effects. Low, buzzy synthesizer tones work particularly well.

**Program Bank 2 (P2)****0.0 StereoLoPass****ADJUST: CutOff 0–127**

This effect is a combination of a stereo low pass filter and a stereo reverb, with the two effects in parallel. Use ADJUST to set the cut off frequency of the filter. The reverb mix is turned all the way down — you can add reverb by changing the FX Mix parameter in the Soft Row, where you'll also find additional parameters for delay voices, and image.

**EQ Effects****(P2 0.0 – P2 1.0)****0.1 Low Pass HiQ****ADJUST: Cutoff 0–127**

The same basic effect as StereoLoPass, but with feedback adjusted to produce a more resonant filter. ADJUST sets the cutoff frequency of the filter.

**0.2 Low Pass LFO****ADJUST: Depth 0–127**

The cutoff of the stereo low pass filter is swept with a rhythmic LFO sine wave. Press **Tap** to synchronize the sweep with rhythmic material. ADJUST controls the low frequency limit of the sweep.

**0.3 StereoHiPass****ADJUST: Cutoff 0–127**

A combination of a stereo high pass filter and a stereo reverb, with the two effects in parallel. Use ADJUST to set the cutoff frequency of the filter. The reverb mix is turned all the way down — you can add reverb by changing the FX Mix parameter in the Soft Row, where you'll also find additional parameters for delay voices, and image.

**0.4 StereoB-Pass****ADJUST: Center 0–127**

A combination of a stereo band pass filter and a stereo reverb, with the two effects in parallel. Use ADJUST to set the center frequency of the filter. The reverb mix is turned all the way down — you can add reverb by changing the FX Mix parameter in the Soft Row, where you'll also find additional parameters for delay voices, and image.

**0.5 Stereo Notch****ADJUST: Center 0–127**

A variation of StereoB-Pass which combines a phase inverted output of the band pass filter with unprocessed signal to create a notch filter. ADJUST sets the center frequency of the notch.

**0.6 SweptNotches****ADJUST: Rate 0–100**

This preset produces vowel-like sounds by sweeping two pairs of notches with two independent time switches. This effect will produce very unusual tonal variations from broad band sources (rich pads, drums, industrial sound effects, etc.) It also generates a good deal of spatial movement within a stereo or Surround mix. ADJUST controls the rate of the sweeps.

**0.7 Env Notches****ADJUST: Release 0–100**

This preset will produce vocalization effects from dynamic sources. The inputs are summed to mono, and the notch filters which are tuned to vocal formants are swept by the resulting input envelope. ADJUST controls the release rate of the input envelope.

**EQ Effects**  
(P2 0.0 – P2 1.0) cont'd.

- 0.8 BandReject 4** **ADJUST: FX Mix 0–100**  
Three independent modulators are used to sweep filters and pan the outputs of this preset. The result is an effect with constantly changing tonal and spatial characteristics. A touch of reverb adds some overall ambience. Use ADJUST to set the mix of filters and reverb.
- 0.9 WaaPedalEko** **ADJUST: FeedBack 0–10**  
Wah-wah with echoes. Foot pedal is patched to filter cutoff. ADJUST controls echo feedback.
- 1.0 RotorWaa** **ADJUST: Rate 0–10**  
A tempo controlled wah-wah effect. ADJUST controls the rhythm of the modulator.

**Spatial Effects**  
(P2 1.1 – P2 2.4)

- 1.1 Movable Echo** **ADJUST: Position 0–127**  
This preset is designed for use with either stereo or Surround mixes. It is a dual delay effect that can be positioned anywhere between the center, side and rear channels. ADJUST sets the position. 0 = center, 64 = side channels, 127 = rear channel.
- 1.2 Movable Hall** **ADJUST: Position 0–127**  
Like Movable Echo, this preset is designed for use with either stereo or Surround mixes. It is a concert hall reverb effect that can be positioned anywhere between the center, side and rear channels. ADJUST sets the position. 0 = center, 64 = side channels, 127 = rear channel.
- 1.3 Go Away** **ADJUST: How Far? 0–127**  
Use this preset to take a stereo source and move it anywhere from in your face to far, far away. Use ADJUST to move the source away from you. 0 = unprocessed stereo, 127 = far away. As ADJUST is increased, EQ, reverb and stereo width are all changed to produce the composite effect. This effect is compatible with stereo and Surround mixes.
- 1.4 Circular Pan** **ADJUST: Rate 0–100**  
Similar to Go Away, this preset sums the inputs to mono and uses the LFO to control EQ, reverb and stereo width to pan the sound through a circular orbit. From right through center, to left through rear, to right. As the sound approaches the center, it grows brighter, louder and dryer. As it recedes towards the rear, it grows darker, softer and more reverberant. In a Surround mix, the center and rear positions of the orbit will feed only the center and rear channels of the mix. ADJUST controls the rate of the pan.
- 1.5 Spatial Hall** **ADJUST: Position 0–35**  
A multi-dimensional preset which changes as you increase or decrease the ADJUST knob value. This preset is not mono compatible.
- 1.6 Rear of Hall** **ADJUST: Decay 0–60**  
This is a really huge space and you're in the back of it. ADJUST changes the reverberation characteristics, making it boomier.
- 1.7 Backstage** **ADJUST: How Far? 0–127**  
Remember what the concert sounded like from the green room? If you need to re-create that ambience, this preset will take you there and let you wander around. Turn up ADJUST to travel further from the stage until, finally, you're out in the parking lot. The Soft Row provides reverb parameters that will let you fine tune the hall to fit your needs.
- 1.8 Steered Rear** **ADJUST: Decay 0–10**  
The Width parameter steers this plate effect from front to back after the AR releases based on input level. ADJUST controls reverb decay. Due to the strong spatial positioning at the end of the decay, this preset is not mono compatible.

- 1.9 Too Deep!** **ADJUST: Decay 0–40** **Spatial Effects (P2 1.1 – P2 2.4) cont'd.**  
The left and right envelope followers control the post delay glides while the AR generator controls reverb width. ADJUST controls reverb decay.
- 2.0 Dyna-Hall** **ADJUST: Decay 0–130**  
A tamer version of Too Deep! No envelope chorusing. Good, beautiful, straight ahead, and spacious.
- 2.1 RotoRox** **ADJUST: FX Mix 0–100**  
RotoRox crossfades deeper into two delay voices when the envelope follower detects an absence of input. Great vocal effects for thickly-produced rock and roll. ADJUST will add more or less reverb into the effects mix.
- 2.2 RotoRoomVox** **ADJUST: Decay 0–100**  
This medium large sized room continuously changes spatial width according to the speed of the LFO which tracks the input. Faster with less input, slower with more input. ADJUST controls reverb decay. Nice roomy effect for background vocals.
- 2.3 RotoRoom** **ADJUST: Decay 0–60**  
Similar to RotoRoomVox except the speed of the spatial width modulation is stable, with the speed of the LFO constant. Room is smaller, more dense-sounding and brighter.
- 2.4 RotoRoom#2** **ADJUST: Rt & Rate 0–60**  
Speed and reverb decay are tied to ADJUST. Way cool on a stinger sound effect! Dynamically spacious and not mono compatible.
- 2.5 Tremolo Tap1** **ADJUST: Depth 0–127** **Gain Effects (P2 2.5 – P2 4.0)**  
This is a basic tremolo effect with a small amount of ambience added. The effect is true stereo. The left and right channels are processed separately to maintain the image of stereo source material. The tremolo rate is tempo controlled (2 cycles/beat). Press **Tap** to synchronize the effect with rhythmic material. ADJUST sets the tremolo depth. 0 = no tremolo, 127 = maximum tremolo.
- 2.6 X-TremoloTap** **ADJUST: Depth 0–127**  
This a variation of TremoloTap1. In this preset the left and right modulation are 90° out of phase, producing a stereo effect that doesn't collapse in a mono mix. If the input source is mono, the sound will seem to move from side to side. If the input source is stereo (or two different mono sources) the left and right channels will alternately grow loud and soft. The modulation rate is tempo controlled. ADJUST controls the tremolo depth for both channels. 0 = no tremolo, 127 = maximum tremolo.
- 2.7 Panner BPM** **ADJUST: Pan Phase 0–1**  
This effect is a tempo controlled auto panner with the inputs independently panned left and right. ADJUST allows you to set the relative phase of the input panning. 0 = in phase, 1 = 90° out of phase. In the preset, the panning is 90° out of phase. If the input source is mono it will pan from side to side. If the input source is stereo (or two different mono sources), the two sources will chase each other between the speakers. Set ADJUST to 0 for "normal" auto panning.
- 2.8 Nice Pan!** **ADJUST: FX Mix 0–100**  
This preset combines static time based delays which move back and forth in the stereo field with a short, bright chorus plate. ADJUST varies the delay and reverb mixing. Great for acoustic guitars.

**Gain Effects**  
(P2 2.5 – P2 4.0) cont'd.

**2.9 Spin & Duck** **ADJUST: Spin Rate 0–127**

In this preset, panning is combined with 6-voice chorus delays and reverb to produce a rich spacious effect. ADJUST controls the panning rate. 0 = slow, 127 = fast. The delays are tempo controlled and ducked by input level. They won't be heard during active passages, but will fade up in the spaces between phrases.

**3.0 MultiFxFade** **ADJUST: FX Mix 0–100**

In this preset, the AR envelope is used to create a moderate fade-in of chorus, delay and reverb effects. ADJUST controls the mix of chorus/delay and reverb. The Soft Row has all the essentials for tweaking the three basic effects, as well as the AR envelope parameters so you can fine tune the fade in rate and threshold. Try this with guitars and keys.

**3.1 Ghost** **ADJUST: FadeShape 0–100**

Source material is accompanied by a ghostly image of itself. Not quite reverb, not quite backwards audio. ADJUST controls the contour of the fade in of the effect. This preset works well with short percussive sources as well as more sustained ones.

**3.2 Ghost Flange** **ADJUST: FadeShape 0–100**

This is a variation of Ghost with some flanging added. Try this with lead guitar. ADJUST controls the contour of the fade in.

**3.3 GhostVibrato** **ADJUST: FadeShape 0–100**

Another Ghost variation. This one will produce a delayed vibrato on sustained notes. ADJUST controls the contour of the fade-in.

**3.4 AutoFadeIn 1** **ADJUST: FX Mix 0–127**

This preset senses input level to produce an automatic volume swell into a chorus/delay reverb effect. Sounds great with guitar and keyboard chords. The fade in rate is moderate. ADJUST controls the mix of chorus/delay and reverb effects. The Soft Row includes parameters for all three effects as well as the envelope parameters which set the characteristics of the fade-in.

**3.5 AutoFadeIn 2** **ADJUST: Fade Rate 0–100**

This variation of AutoFadeIn allows you to set the fade-in rate with ADJUST.

**3.6 AutoFadeIn 3** **ADJUST: Rvb Lvl 0–127**

This variation of AutoFadeIn 1 has shorter delay times and a lighter touch on the chorus effect. ADJUST sets the mix of chorus/delay and reverb effects.

**3.7 ChordSwells** **ADJUST: Rvb Lvl 0–127**

A more dramatic version of AutoFadeIn. The fade time is quite long, and the chorus and delay effects are fairly strong. Try it with sustained piano or guitar chords. ADJUST controls the mix of chorus/delay and reverb.

**3.8 BowedChords** **ADJUST: FX Mix 0–100**

A more subtle version of AutoFadeIn. Chorusing is turned off and delays are more subdued. This is essentially a volume swell into reverb. Use it to pull cello-like tones from sustained notes or block chords. Very nice with grand piano. ADJUST sets the mix of clean echoes and reverb.

**3.9 BowedEchoes** **ADJUST: FX Mix 0–100**

In this preset, input level controls echo feedback as well as triggering the fade-in. Feedback is reduced when the signal is loud and turned up when the signal fades out. This produces echoes with soft attacks that appear when the input signal drops. ADJUST controls the mix of echoes and reverb.

- 4.0 Pedal Swell** **ADJUST: FX Mix 0–100** **Gain Effects (P2 2.5 – P2 4.0) cont'd.**  
 This is a combination of four 400 ms delays, a slight amount of chorus, and just a hint of reverb. The foot controller is controlling the left and right input levels which allows you to get majestic volume swells. ADJUST controls the combination of delays and chorus with reverb.
- 4.1 Rez Climber** **ADJUST: Tuning 0–60** **Resonant Chord Effects (P2 4.1 – P2 4.9)**  
 Try this preset with dialog or unpitched source material. It is a special effect that dynamically sweeps the resonators through the 2nd, 3rd, 4th and 5th pitches of the harmonic series. The sweep is controlled by the AR Env, which is triggered by mono input level. ADJUST sets the fundamental pitch for the series.
- 4.2 Chord Walk** **ADJUST: High Cut 0–50**  
 This preset uses the resonators to generate a rhythmic chord pattern from unpitched source material. ADJUST controls the high frequency content of the chords. The time switches, Sw 1 and Sw 2 are used to change the chord root-note and mode and also vary the rhythm. Listen to this effect with a simple kick, snare and hi-hat pattern as an input source. Press **Tap** twice to synchronize the tempo of the effect with the tempo of the drum pattern.
- 4.3 Mars Bars** **ADJUST: High Cut 0–50**  
 In this preset, the resonators are used to create a truly weird and spacey special effect. Use any continuous sound effect, dialog, or even instrumental tracks as source material. The tuning of the resonators is swept slowly through a series of pitches by the LFO. The resulting sound is quite unusual (an orchestra from another planet?). ADJUST controls the overall high frequency content of the effect. Be sure to check out the Soft Row, which contains parameters for tuning, voice assignment and more.
- 4.4 Modallmpulse** **ADJUST: Tone 0–50**  
 The resonators in this preset are tuned to arpeggiate a modal 7th chord. ADJUST changes the overall tone of the arpeggiated notes. The rhythm of the arpeggio is tempo controlled. Press **Tap** twice to synchronize it to the beat. The effect works well with single percussion hits (try it with kick or snare). You can change the key, scale, and root note of the arpeggio by playing with the Pitch sub parameters in the Soft Row. You'll also find controls for reverb and delay there as well.
- 4.5 Major Minor** **ADJUST: Tone 0–50**  
 This effect in this preset builds a 6-note modal chord one note at a time. Use a single percussion hit as an input source (a slowly repeating snare hit works well). The notes in the chord are added slowly over 24 beats. The effect is tempo controlled. Press **Tap** twice to synchronize it to the beat. ADJUST changes the overall tone of the chord notes. As the chord builds, it is changed rhythmically between major and minor scales. The root note of the chord is also changed in a rhythmic manner. The LFO controls the rate of the build and the major minor shift. Switch 2 controls the changing root notes.
- 4.6 MIDiChords** **ADJUST: Sparkle 0–5**  
 This preset is driven by MIDI Note Number. Resonators follow the note in a diatonic tone cluster. ADJUST controls the brightness of the resonators. If the audio source is the same as the MIDI source (a synthesizer), the effect is a little like harmonization. If the audio source is different, the effect is much harder to describe. This preset will pass audio if there is no MIDI input.

**Resonant Chord Effects  
(P2 4.1 – P2 4.9) cont'd.**

- 4.7 LvlSweeper** **ADJUST: Color 0–24**  
This preset is driven by level. Any input that exceeds the input threshold will cause a little burst of resonators that quickly swirl through the stereo field. ADJUST controls the pitch range of the resonators.
- 4.8 Sweeper** **ADJUST: Pitch 0–24**  
This preset is a resonant arpeggiator in which a tempo-driven LFO controls the resonator pitches. ADJUST controls the pitch range of the arpeggio effect.
- 4.9 MIDISustain** **ADJUST: Bright 0–5**  
This preset requires MIDI note input or it will not pass audio. Resonators are assigned as the notes are played (when playing chords, it's best to spread them a little). Footswitch 2 works like a piano damper pedal. When the audio source is the same as the MIDI source, the effect is a little like playing a piano while holding down the pedal. When the audio source is different, well ... ADJUST controls the brightness of the resonators.

**Program Bank 3 (P3)****Reverb Effects  
(P3 0.0 – P3 2.8)**

- 0.0 Small+Stage** **ADJUST: Stage Lvl 0–50**  
This preset combines a smooth, small reverberant space with several stage reflections. Use ADJUST to set the stage reflection level.
- 0.1 Small Room** **ADJUST: Liveness 0–60**  
Use ADJUST to quickly change the ambient characteristics of this typically tight sounding room. Great for ADR work.
- 0.2 Living Room** **ADJUST: Decay 0–60**  
Another ADR preset, medium spaced but with a short Rt. ADJUST quickly changes if not perfect for your application.
- 0.3 Brick Kick** **ADJUST: Liveness 0–100**  
Kicks \*ss on kick drums or the entire submix. ADJUST increases the liveness of the chamber.
- 0.4 Large Room** **ADJUST: Decay 0–60**  
A perfectly smooth listening room with medium diffusion. Very natural sounding on any sound source. ADJUST lengthens or shortens the reverb decay.
- 0.5 SnareChamber** **ADJUST: Liveness 0–60**  
Like Brick Kick, ADJUST increases the liveness of the space. A classic from the PCM 70.
- 0.6 Tiled Room** **ADJUST: Decay 0–60**  
A future classic... ADJUST controls reverb decay.
- 0.7 Rich Chamber** **ADJUST: Decay 0–60**  
Smooth and full sounding. ADJUST varies reverb decay.
- 0.8 Vox Chamber** **ADJUST: Liveness 0–60**  
Combines recirculating echoes which fall away quickly once signal is absent. Increasing ADJUST lengthens the reverb decay, which will mask the reflection echoes. For vocals.
- 0.9 Locker Room** **ADJUST: Depth 0–127**  
Just like in high school. ADJUST creates a deeper, more reverberant sound.
- 1.0 Wide Chamber** **ADJUST: Decay 0–60**  
Big and wide sounding with a preset short Mid Rt. ADJUST will change that if necessary. Use on synth pads or vocals.

**ADJUST:      Duration      0-64**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     **Decay**            **0-30**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:    Decay            0-60**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     **Decay**             **0–60**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     Decay            0-60

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     Decay             0-60

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:     Decat             0-60**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:** Wave Knob 0–40

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:    Decay            0-100**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

## ADJUST: Emty/Full 0-15

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     Decay             0-60

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     Decay             0-60

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:**     Decay            0-60

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.

**ADJUST:      FX Mix            0-100**

**Reverb Effects**  
(P3 0.0 – P3 2.8) cont'd.



**Reverb Effects**  
**(P3 0.0 – P3 2.8) cont'd.**

- 2.5 Plate4 Horns** **ADJUST: Impact 0–100**  
 Bright and percussive for those horn tracks needing that certain edge. Use ADJUST to modify the attack and release characteristics.
- 2.6 Alley Slap** **ADJUST: Pre Delay 0–100**  
 Great for any punctuated sound source — vocals, guitars, anything. Use ADJUST to match the delay to the music.
- 2.7 Drum Gate** **ADJUST: Duration 0–70**  
 This inverse gated preset is low on diffusion and high on attitude. Made for drums and other high impact sources. ADJUST controls the duration of the effect.
- 2.8 Slope Down** **ADJUST: Duration 0–100**  
 Very inverse sounding effect. ADJUST varies the length of the slope. The AR Envelope, triggered by mono level, cranks in a downward spiraling pitch shift. Unearthly on voices.

**Processed Reverb Effects**  
**(P3 2.9 – P3 3.9)**

- 2.9 BigBoomRoom** **ADJUST: BoomKnob 0–36**  
 The Envelope followers glide the left and right post delays giving a strong modulation effect to the reverb tail. ADJUST adds more or less boom. Use on punchy low frequency sound sources.
- 3.0 Whammy Hall** **ADJUST: Decay 0–30**  
 If your guitar doesn't have a whammy bar, it does now. The AR generator is triggered from the mono level source. The AR generator is then attached to the LFO depth which drives the two Post Delays after the reverb. The Release constant is set rather long so that the depth reaches full scale after 2.2 seconds. ADJUST sets the reverb time.
- 3.1 JetChamber** **ADJUST: Jet Knob 0–127**  
 A big chamber reverb with stereo flanging on the outputs. Use ADJUST to set the amount of "whoosh." The rate of the flange is controlled by the LFO. You'll find it in the Soft Row along with the essential reverb parameters.
- 3.2 EnvChamber** **ADJUST: Jet Knob 0–127**  
 Similar to JetChamber, but here flanging is controlled by input level. The flange effect is most prominent when the input level drops. This lets you process drums with little or no reverb flanging while they are active, but individual hits and fills will have pronounced flanging during the reverb tail. This also works well with a variety of sources including acoustic guitar and piano.
- 3.3 Wizz and Wazz** **ADJUST: Decay 0–127**  
 This tempo driven effect has delays that whiz from left to right. FX Mix is also linked to tempo, going slowly from delays to reverb, then abruptly back to delays. ADJUST controls reverb decay.
- 3.4 Sci Fi** **ADJUST: Wiggle 0–127**  
 This preset uses an LFO modulated reverb for its basic effect. ADJUST controls the depth of the modulation. It will put a big spacious halo around unpitched sources, like drums. Lower settings of ADJUST will work best with pitched sources, although higher settings can yield some interesting sound effects. (Listen to a high piano note with ADJUST at about 100 —definite science fiction material!)
- 3.5 Wobble Plate** **ADJUST: Wobble 0–30**  
 A different type of chorus plate. ADJUST controls glide response.

**3.6 Dyna Vibrato** **ADJUST: Glide 0–127**  
 Input level triggers a delayed vibrato. The vibrato is created by modulating two pairs of gliding delays. ADJUST controls the offset between the delay pairs. Use it to thicken up the effect. You'll find parameters to change the modulation depth and add reverb in the Soft Row. A nice effect to sweeten up acoustic guitar, dry sampler or synth tracks, etc.

**Processed Reverb Effects  
 (P3 2.9 – P3 3.9) cont'd.**

**3.7 VibroVerb** **ADJUST: V-Depth 0–127**  
 In this effect, the reverb is processed to produce a vibrato that wanders slowly between two rates. ADJUST sets the depth of the vibrato. 0 = none, 127 = maximum. Low settings of ADJUST work nicely to open up the space around backing tracks. Higher settings can be used to add character to dry synthesizer tones, guitar and piano. Check out the Soft Row for parameters that you use to change the overall sound of the reverb.

**3.8 SweepVerb** **ADJUST: GldResp 0–100**  
 The left and right reverb outputs are detuned in opposite directions by the AR, which is triggered by input level. ADJUST controls the amount of detune. To open up the space around percussive sources, use very low settings of ADJUST. Higher settings will produce radical pitch swoops in the reverb. The Soft Row contains several useful reverb parameters.

**3.9 EnveloVerb** **ADJUST: Rt HC 0–40**  
 Great on sustained and slowly fading sound sources. The AR generator controls the Reverb output as well as the Reverb time. ADJUST controls the high frequency response of the reverb tail.

**4.0 Super Ball!** **ADJUST: Bounce It 0–1**  
 This preset will let you turn any source into a bouncing ball. Run a track or instrument, click ADJUST to 1 and back to 0 and sit back. The effect captures a stereo sample on the fly, loops it and decreases the loop size as it repeats. (Reverb is added while the loop plays.) When the loop size gets to zero, the effect resets and starts sampling the inputs again. The loop size is tempo controlled. Press **Tap** to synchronize it with rhythmic material.

**Remix Effects  
 (P3 4.0 – P3 4.9)**

**4.1 LevlStutter** **ADJUST: Mstr Fbk 0–100**  
 Delays are driven by input level (high level = shorter delays). The delays are also driven by tempo. Works best for deconstructing dialog, but is also interesting with any audio with lots of holes.

**4.2 Freeze 2 of 8** **ADJUST: Feedback 0–100**  
 This stereo preset automatically freezes 2 beats out of every 8. Press **Tap** to synchronize the effect with the source material. ADJUST lets you set the feedback level when the loop is off. Be sure to try this one with a dance mix, or MIDI drum patterns.

**4.3 Freeze 2over3** **ADJUST: Feedback 0–100**  
 A slightly more diabolical version of Freeze 2 of 8. This preset also loops 2 out of every 8 beats, but the left and right delays are set to different rhythmic values. The result is a 2 against 3 pattern that alternates between the left and right channels. ADJUST sets the feedback for both delays while the loop is off.

**4.4 Freeze&Speed** **ADJUST: Freeze It 0–1**  
 This preset allows you to grab and freeze portions of the source material on the fly. Click ADJUST from 0 to 1 to freeze the loop. It will repeat infinitely, but will grow more diffused as it repeats. Click ADJUST back to 0 to turn the loop off. The loop will speed up for a short time before turning off. The AR Release parameter (found in the Soft Row with other goodies) sets the speed-up time. The loop size is tempo controlled. You can change it on the fly by pressing **Tap** while the loop is off.

**Remix Effects**  
(P3 4.0 – P3 4.9) cont'd.

**4.5 ADJToFreeze** **ADJUST: Freeze It** **0–1**  
Click ADJUST to create an infinite stereo loop of the input source. Click it again to stop the loop. A little reverb is added while the loop is on. The loop size is tempo controlled. Change it on the fly by pressing **Tap** while the loop is off. The preset is set to freeze 1 beat, but you can set any rhythm you want by changing the Soft Row delay parameter values.

**4.6 ADJToFreeze2** **ADJUST: Freeze It** **0–1**  
The same basic loop effect as ADJToFreeze, but the delays are set to different values for the left and the right side. In this preset a 2 against 3 pattern will be created.

**4.7 FS2 Freeze** **ADJUST: Feedback** **0–100**  
A very useful infinite delay effect. Press **Tap** to establish the tempo. Use ADJUST to set the feedback level. Foot switch 2 is patched to turn infinite repeat on and off.

**4.8 FS2 Speedup** **ADJUST: How Long?** **0–100**  
Footswitch 2 is patched to turn on an infinite loop while it is held down. When the foot switch is released, the loop continues but grows shorter with every repeat until the loop size is 0. Use ADJUST to set the amount of time it takes for the loop to shrink to nothing. *A footswitch must be connected to make this preset work.*

**4.9 LongestLoop** **ADJUST: FX Mix** **0–100**  
This preset uses the cross-feedback path in the delay voices to create a 5-second mono loop. Footswitch 2 is patched to turn infinite repeat off and on and, at the same time, to change the mix from 0 % to 100 % wet. Foot switch 1 is patched to clear the delays. ADJUST sets the reverb level for the loop. 0 = no reverb, 127 = all reverb. The Soft Row contains the parameters for setting the master delay for the loop as well as several reverb parameters. *A footswitch must be connected to make this preset work.*

**Program Bank 4 (P4)**

NOTE: Most of the Pitch presets in Banks 4 and 5 are set to 100% wet. To hear the harmony/detune in many presets, dry signal should be added from the mixer. If the PCM 81 is connected in line, adjust the Pitch FX Mix parameter (**Control 0.0**) to add some dry signal.

**Vocal Shift**  
(P4 0.0 – P4 0.9)

**0.0 FixYerVox** **ADJUST: Offset** **0–100**  
Turns your marginal singer into a rock'n'roll star. Bounce your vocals to another track and fix the pitch on the fly. Stereo In/Stereo Out. ADJUST gives you  $\pm 100$  cents of offset to correct any other intonation.

**0.1 FixYerVoxChm** **ADJUST: Offset** **0–100**  
Similar to FixYerVox with an added vocal chamber for occasions when you run out of tracks and need to fix the pitch while you mix. Stereo In/Stereo Out.

**0.2 ThickFixVox** **ADJUST: Thickness** **0–50**  
Similar to FixYerVox with stereo detuning for occasions when you run out of tracks and need to fix the pitch while you mix. Stereo In/Stereo Out. ADJUST controls the amount of detuning, in cents.

**0.3 Cocomo** **ADJUST: Thickness** **0–100**  
The inputs are detuned. The left is delayed 31ms, the right is delayed 47ms. ADJUST controls the amount of + and – detuning in cents. Stereo In/Stereo Out.

**0.4 QuadDTune** **ADJUST: Tune Knob** **0–100**  
Fatten up vocals with four pitch shift voices panned across the stereo field. ADJUST sets the amount of detuning. (Two voices go sharp, two go flat.)

- 0.5 ADTStereoIn** **ADJUST: LFO Rate 0–100** **Vocal Shift**  
 The left input is pitch shifted up, the right input is shifted down. An LFO modulates the pitch from 2-12 cents. Stereo In/Stereo Out. ADJUST controls the LFO rate from 0-25Hz. **(P4 0.0 – P4 0.9) cont'd.**
- 0.6 TripleTrack** **ADJUST: Separate 0–100**  
 Adds two vocal tracks. ADJUST controls how closely the pitch shifted voices track the pitch and timing of the input track.
- 0.7 Quad Track** **ADJUST: Separate 0–100**  
 Similar to TripleTrack, this preset adds a third doubling to the original track. ADJUST controls how tightly the three shifted voices track the original.
- 0.8 Quint Track** **ADJUST: Separate 0–100**  
 Makes one voice sound like five by adding four double tracks to the original. Each one is slightly out of time and out of tune with the others. ADJUST controls how tightly the shifted voices track the original.
- 0.9 Add Voices** **ADJUST: How many? 1–4**  
 A variation of QuintTrack with ADJUST adding more doublings to the original. 1=1 voice with slight detune, 2=2 voices, 3=3 voices, 4=4 voices.
- 1.0 Major Thirds** **ADJUST: Inversion 0–4** **Vocal Harmony**  
 This preset allows you to treat the input note as the root or third in major third harmony. ADJUST lets you select the inversion. 0=slight detune, 1=sixth below, 2=third below, 3=third above, 4=sixth above. **(P4 1.0 – P4 1.9)**
- 1.1 Minor Thirds** **ADJUST: Inversion 0–4**  
 This preset allows you to treat the input note as the root or third in minor third harmony. ADJUST lets you select the inversion. 0=slight detune, 1=sixth below, 2=third below, 3=third above, 4=sixth above.
- 1.2 Major Triads** **ADJUST: Inversion 0–6**  
 The input note can be harmonized as the root, third or fifth of a major chord. ADJUST selects the harmony and inversion. 0=slight detune, 1=input is root (1st inversion), 2=input is root (2nd inversion), 3=input is third (1st inversion), 4=input is third (2nd inversion), 5=input is fifth (1st inversion), 6=input is fifth (2nd inversion).
- 1.3 Minor Triads** **ADJUST: Inversion 0–6**  
 The input note can be harmonized as the root, third or fifth of a minor chord. ADJUST selects the harmony and inversion. 0=slight detune, 1=input is root (1st inversion), 2=input is root (2nd inversion), 3=input is third (1st inversion), 4=input is third (2nd inversion), 5=input is fifth (1st inversion), 6=input is fifth (2nd inversion).
- 1.4 Major & Dbls** **ADJUST: Inversion 0–4**  
 The input note is doubled and also harmonized as the root of a major triad. ADJUST selects the inversion of the chord.
- 1.5 Minor & Dbls** **ADJUST: Inversion 0–4**  
 The input note is doubled and also harmonized as the root of a minor triad. ADJUST selects the inversion of the chord.
- 1.6 4ths & Dbl** **ADJUST: Inversion 0–4**  
 The input note is doubled and also harmonized as the root of chord made of four perfect fourths. ADJUST selects the inversion of the chord.

**Vocal Harmony**  
(P4 1.0 – P4 1.9) cont'd.

**1.7 MIDI Pitch** **ADJUST: Rvb Mix 0–100**  
Use this preset to harmonize source material via single notes played on a MIDI keyboard. The amount of pitch shift is determined by the size of the interval played above or below middle C. ADJUST allows you to mix in some reverberation. Note that the audio source should not be the same source used to generate the MIDI notes.

**1.8 MIDI Harmony** **ADJUST: Rvb Mix 0–100**  
This is a useful variation of MIDI Pitch. Play two notes at once on a MIDI keyboard to add two voices to the input note to create three part harmony. The harmonies are relative to middle C. ADJUST allows you to mix in some reverb. Note that the audio source should not be the same source used to generate the MIDI notes.

**1.9 MIDI Chord** **ADJUST: V2 Pitch 1–25**  
A variation of MIDI Pitch that allows you to create four part harmonies. You can add two notes to the input note from a MIDI keyboard and dial-in an additional fixed harmony with ADJUST which provides 1 octave up or down in half steps. Note that the audio source should not be the same source used to generate the MIDI notes.

**Guitar Magic**  
(P4 2.0 – P4 2.9)

**2.0 SingleDetune** **ADJUST: 10=unison 0–20**  
A high quality detuner. ADJUST shifts pitch up or down 10 cents. Detune is at unison when ADJUST=10.

**2.1 DualDetune** **ADJUST: cents +/- 0–100**  
A classic guitar effect. Two pitch shifters are used to spread out and thicken up the sound. ADJUST controls the amount of shift in cents. One side shifts up, the other shifts down.

**2.2 Phat Detune** **ADJUST: TuneKnob 0–100**  
A classic detune effect with two additional voices. The four voices are panned across stereo space to really widen up the sound. ADJUST controls the overall amount of detuning.

**2.3 EGtrPitchRm** **ADJUST: Fbk Mix 0–50**  
This preset produces a rich, room-like halo around every note. Sounds like reverb, but it's not. How can you tell? Choke off a note and you'll hear that there's no decay. The effect is created by using pitch shifted echoes with cross feedback. The input level controls the amount of cross feedback. When the input dies out, the X-feedback goes to 0. ADJUST lets you dial some back in.

**2.4 VibroShift1** **ADJUST: Rate 0–100**  
A simple unison vibrato with ADJUST controlling the rate. Mix with the original signal for chorusing or use 100% wet for straight vibrato.

**2.5 VibroShift2** **ADJUST: Rate 0–100**  
Similar to VibroShift1 with a touch of octave down vibrato. Nice on single notes or chords.

**2.6 VibroTrem** **ADJUST: Rate 0–100**  
Pitch shift vibrato and out-of-phase tremolo combined to create a new take on a retro sound. ADJUST controls the rate of the throb. For the fullest effect, don't mix any dry signal with the PCM 81 output.

**2.7 VibroTremRvb** **ADJUST: Rate 0–100**  
Another cool take on a vintage effect with reverb added between the vibrato and the tremolo. Play some chunky chords and listen to the verb pump. ADJUST controls the rate.

- 2.8 OctDown Verb** **ADJUST: Oct Mix 0–100** **Guitar Magic (P4 2.0 – P4 2.9) cont'd.**  
Great with muted single note licks and runs, this preset also works well with chords. The reverb is fed by a unison vibrato and an additional vibrato shifted down one octave. ADJUST controls the mix of the unison and octave. 0=unison only, 50=unison and octave down, 100=octave down only.
- 2.9 PitchSlapRvb** **ADJUST: Rvb Tune 0–100**  
This preset delivers a tight reverb slap after each note. The trick here is that you can tune the reverb sharp relative to the original note. ADJUST controls the reverb tuning — a little goes a long way.
- 3.0 24 String** **ADJUST: 8va Level 0–100** **Instrument Shift & Harmony (P4 3.0 – P4 3.9)**  
Fatten up your 6-string with both detuning and an octave up. Stereo In/Stereo Out.
- 3.1 HonkyTonkPno** **ADJUST: Delay 0–100**  
Go West, young man, to the frontier...and don't forget the pretzels and beer. Stereo In/Stereo Out.
- 3.2 PowerNotes** **ADJUST: 5ths Lvl 0–100**  
This preset is tuned up to give you a power chord when you play a single note. Each note is doubled at the unison and at an octave down. ADJUST lets you tune in two fifths — one below and one above the input note.
- 3.3 4-NoteChords** **ADJUST: Inversion 0–6**  
This preset treats the input note as the root or fifth of a major seventh, minor seventh or dominant seventh chord. ADJUST lets you select the chord and the inversion.
- 3.4 ModalChords1** **ADJUST: Mode 1–8**  
This preset will produce four-note chords relative to the input note. ADJUST selects the chord. The chords are voiced in parallel, each chord in root position.
- 3.5 ModalChords2** **ADJUST: Mode 1–8**  
A variation of ModalChords1 with the chords voice-led in close position to minimize the amount of pitch shifting required to produce each chord.
- 3.6 Chromatic Up** **ADJUST: HalfSteps 0–12**  
Straightforward and simple, ADJUST lets you tune any chromatic interval up to an octave above the input note.
- 3.7 Chromatic Dn** **ADJUST: HalfSteps 0–12**  
This variation lets you dial in any chromatic interval down to an octave below the input note.
- 3.8 FootPdl Oct** **ADJUST: Up / Down 0–1**  
Designed to be used with a foot pedal. When connected, it produces a whammy bar slide up or down one octave. ADJUST sets the direction of the slide. 0=up, 1=down.
- 3.9 FootPdlChord** **ADJUST: UnisonMix 0–100**  
Designed to be used with a foot pedal. In this case, the pedal slides two voices from unison to a major sixth (a fourth below and major third above the input note). ADJUST allows you to mix in the original note so you can pedal-slide major chords.

### Percussion & Tempo (P4 4.0 – P4 4.9)

- |                                                                                                                                                                                                                                             |                        |              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|
| <b>4.0 SnareTools 1</b>                                                                                                                                                                                                                     | <b>ADJUST: Crack</b>   | <b>0–100</b> |
| Natural snare reverb with ADJUST providing “crack” control. Detune is added to the original input before both are sent to the reverb. FX Mix controls the blend of dry vs. pitched signal to achieve optimal “crack”. Stereo In/Stereo Out. |                        |              |
| <b>4.1 SnareTools 2</b>                                                                                                                                                                                                                     | <b>ADJUST: Crack</b>   | <b>0–100</b> |
| Trippy backwards effect. Reverb feeds the pitch shifter. Mono In/Stereo Out.                                                                                                                                                                |                        |              |
| <b>4.2 SnareTools 3</b>                                                                                                                                                                                                                     | <b>ADJUST: Crack</b>   | <b>0–100</b> |
| Inverse snare reverb with “crack” control via ADJUST. Detune is added to the original input before both are sent to the reverb. FX Mix controls the blend of dry vs. pitched signal to achieve optimal “crack”. Mono In/Stereo Out.         |                        |              |
| <b>4.3 TomTomTools</b>                                                                                                                                                                                                                      | <b>ADJUST: Boom!</b>   | <b>0–100</b> |
| Turns oatmeal boxes into cannons. Stereo In/Stereo Out.                                                                                                                                                                                     |                        |              |
| <b>4.4 Slap Up BPM</b>                                                                                                                                                                                                                      | <b>ADJUST: pitch</b>   | <b>0–100</b> |
| A tuned percussion plate with a slap predelay. Great for unpitched percussion. Use ADJUST to tune the pitch of the reverb above the pitch of the percussion.                                                                                |                        |              |
| <b>4.5 Slap Dn BPM</b>                                                                                                                                                                                                                      | <b>ADJUST: pitch</b>   | <b>0–100</b> |
| A variation with ADJUST tuning the reverb below the pitch of the percussion source.                                                                                                                                                         |                        |              |
| <b>4.6 Thick Slap</b>                                                                                                                                                                                                                       | <b>ADJUST: detune</b>  | <b>0–100</b> |
| This preset detunes the reverb above and below the original source. A great alternative for thickening drum sounds.                                                                                                                         |                        |              |
| <b>4.7 RepeatSlaps</b>                                                                                                                                                                                                                      | <b>ADJUST: Fbk</b>     | <b>0–100</b> |
| Pitched reverb echoes produce arpeggios. Try this with any short percussive track for an interesting effect. ADJUST controls the amount of feedback (decay of the arpeggio).                                                                |                        |              |
| <b>4.8 AutoScratch</b>                                                                                                                                                                                                                      | <b>ADJUST: Rvb Mix</b> | <b>0–100</b> |
| Run a rhythm bed or drum machine into this preset and it will “scratch it” in tempo. ADJUST adds reverb. Press <b>Tap</b> twice to lock in the rhythm.                                                                                      |                        |              |
| <b>4.9 Stop n Go</b>                                                                                                                                                                                                                        | <b>ADJUST: Rvb Mix</b> | <b>0–100</b> |
| A more radical version of AutoScratch.                                                                                                                                                                                                      |                        |              |

### Program Bank 5 (P5)

- |                                                                                                                                                                                                                                                |                         |                             |              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------|--------------|
| <b>Sci-Fi FX<br/>(P5 0.0 – P5 0.9)</b>                                                                                                                                                                                                         | <b>0.0 PullThePlug!</b> | <b>ADJUST: Off/On</b>       | <b>0–1</b>   |
| Use this special effect to simulate the sound of a tape machine, juke box (etc.) running down when the power has been cut off in the middle of a tune. Turn ADJUST from 1 to 0 to cut the power, and from 0 to 1 to turn it back on.           |                         |                             |              |
|                                                                                                                                                                                                                                                | <b>0.1 Stargate</b>     | <b>ADJUST: Velocity</b>     | <b>0–100</b> |
| A dramatic special effect that transforms a single percussive sound into a spatial wash of ascending and descending pitches. Try this with single drum hits or short sound effect samples. ADJUST controls the rate of the ascent/descent.     |                         |                             |              |
|                                                                                                                                                                                                                                                | <b>0.2 WhiteHole</b>    | <b>ADJUST: Big&gt;Small</b> | <b>0–100</b> |
| A variation of Stargate that transforms a single sonic event into a randomized series of pitched echoes. ADJUST controls the pitch and delay deviation.                                                                                        |                         |                             |              |
|                                                                                                                                                                                                                                                | <b>0.3 MotherShip</b>   | <b>ADJUST: Velocity</b>     | <b>0–100</b> |
| Need some processing for lift-off or acceleration effects? This effect delivers. Drive it with a single impulse or short sound effect to produce a spatial glide that ascends to the limits of audibility. ADJUST controls the rate of ascent. |                         |                             |              |

#### 0.4 XplodeDown

**ADJUST:**    Velocity    0-100

Use this preset to add more drama to special effects (or even single drum hits). This preset transforms the input source into an eerie downward glide with ADJUST controlling the rate of the glide.

**Sci-Fi FX**

(P5 0.0 – P5 0.9) cont'd.

## 0.5 Xplode Up

**ADJUST:**    **Velocity**        **0-100**

A variation of XplodeDown with the input source transformed into an upward glide.

## 0.6 DialogCloner

**ADJUST:**    Tuning            0–100

A special effect for speech, this preset creates the effect of several people speaking at once. Four additional voices are added above and below the pitch of the input source. ADJUST controls the tuning spread of the four clones.

## 0.7 Toon Voices

**ADJUST: Character 0-7**

Changes the character of vocals for special effect dialog tracks. ADJUST selects several different pitch shift amounts to create a range of characters from scary monsters, to chipmunks, to talking bees.

## 0.8 Evil Voice

**ADJUST: Pitch 0-100**

Downward pitch shift and a touch of reverb are used to turn a normal male speaking voice into something decidedly dark and nasty. ADJUST allows you to dial the appropriate touch of evil.

## 0.9 Evil Echoes

**ADJUST: Pitch 1-100**

This special effect has echoes that change pitch as they repeat. ADJUST controls the amount of shift per repeat. 1-49=descending pitch, 50=no pitch change, 51-100=ascending pitch.

## 1.0 Pitch Across

**ADJUST:**     **Detune**            **0–100**

A single note produces a 4-voice panned delay. ADJUST controls the amount of pitch shift for the four voices. Small values produce detuning. The maximum value (100) produces a major arpeggio. The delay rhythm is tempo-controlled. Press **Tap** twice to lock in with the beat.

## Pitch & Delay

(P5 1.0 – P5 1.9)

### 1.1 PitchEkoRvb

**ADJUST:    Eko/Rvb    0-100**

The dual pitch shifters produce a detune/echo effect with reverb. ADJUST lets you set the balance of the two effects. 0=detune echoes only, 50=detune echoes and reverb, 100=reverb only.

## 1.2 PitchPong

**ADJUST:**     DelayRate     0–100

The inputs are delayed, pitched down a little, then cross fed to the other side. Slap echoes bounce from left to right and fall slightly in pitch. The LFO adds chorusing to the pitch change. Stereo In/Stereo Out.

### 1.3 AR DlyVerb

**ADJUST:**     **Detune**            **0–100**

When the input level drops, delays fade away into reverb. Stereo In/Stereo Out.

## 1.4 In The Air

**ADJUST:      Go                      0-1**

The inputs are detuned + and - 8 cents. When ADJUST is changed from 0 to 1, a quarter-note delay recirculates through the detuner. Stereo In/Stereo Out.

### 1.5 Thick>Ducked

**ADJUST:**     **Detune**            **0–100**

The inputs are detuned. When the input level drops below the threshold, quarter-note delays with regeneration recirculates into the detuners. Stereo In/Stereo Out.



**Pitch & Delay**  
**(P5 1.0 – P5 1.9) cont'd.**

- 1.6 FootPdlEkos** **ADJUST: Detune 0–100**  
 Designed to be used with a foot pedal. A detune chorus is routed through stereo delays and reverb. The pedal controls the input level to the delays and reverb. Use the pedal to capture specific phrases in the delays/reverb. Great with guitar and other solo instruments.
- 1.7 DualDTuneDly** **ADJUST: cents +/- 0–100**  
 This preset combines two detuned voices with delays. ADJUST controls the amount of detune. Controls for the delays (and reverb too) are in the Soft row.
- 1.8 QuaDTuneDly1** **ADJUST: TuneKnob 0–100**  
 For thicker detune and delay effects, this preset uses four pitch shifters for detuning. ADJUST controls the pitch spread of the four shifters. Delay and reverb controls are in the Soft row.
- 1.9 QuaDTuneDly2** **ADJUST: TuneKnob 0–100**  
 Similar to QuadTuneDly1 with the delays set to rhythmic values (eighths and triplets). Press **Tap** to synchronize the rhythms to the beat.

**Pitch Sequences**  
**(P5 2.0 – P5 2.9)**

- 2.0 Major Across** **ADJUST: MasterDly 0–100**  
 A major triad is arpeggiated across the stereo field. ADJUST controls the overall timing of the arpeggiation.
- 2.1 Minor Across** **ADJUST: MasterDly 0–100**  
 A minor triad is arpeggiated across the stereo field. ADJUST controls the overall timing of the arpeggiation.
- 2.2 Dim hARP** **ADJUST: GlissRate 0–100**  
 The inputs rise in a diminished arpeggiated scale. This preset turns one harp note into a glissando, or one quarter note into a harp chord. Individual voices are panned across the stereo field. Stereo In/Stereo Out.
- 2.3 Dim hARP Vrb** **ADJUST: GlissRate 0–100**  
 Similar to Dim hARP with an added reverb.
- 2.4 DimScaleDown** **ADJUST: ScaleRate 0–100**  
 The inputs descend into a diminished arpeggiated scale. This preset turns one note into a scale or glissando. Individual voices are panned across the stereo spectrum. A little reverb is added.
- 2.5 GlassCascade** **ADJUST: MasterDly 0–100**  
 A glassy, chime-like tail makes this preset great for sustained single-note melodies. Spooky and pretty at the same time. Stereo In/Stereo Out.
- 2.6 Carnival** **ADJUST: MstrFdbk 0–100**  
 Turns guitar or piano into steel drums, especially if you play diatonic thirds, as steel drummers have been known to do. Stereo In/Stereo Out.
- 2.7 Sequence 1** **ADJUST: MasterFbk 0–100**  
 Play single note lines. Adds M6 and M9. Tap tempo-driven. Stereo In/Stereo Out.
- 2.8 Sequence 2** **ADJUST: MasterFbk 0–100**  
 Similar to Sequence1 with a very different sound.
- 2.9 Pentatonics!** **ADJUST: MasterFbk 0–100**  
 A sequence of notes from the minor pentatonic scale are played for each input note. The input note is treated as the root of the scale. ADJUST controls the amount of feedback.

- 3.0 JurassicSolo** **ADJUST: Decay 0–100** **Pads & Drones (P5 3.0 – P5 3.7)**  
 A very long plate reverb is pitched down a minor third. Spooky. Good for single-note melody soundtracks. Stereo In/Stereo Out.
- 3.1 SwrlWhlnt** **ADJUST: MasterDly 0–100**  
 Swirling Whole Note — An LFO controls FX Width. FX Mix is 100% reverb. Delay time is 0ms. Slightly detuned swirling voices produce a reverb that is especially wonderful on whole notes. Stereo In/Stereo Out.
- 3.2 Sweet Chorus** **ADJUST: Spd/Width 0–100**  
 An LFO sine wave alternates detuning from positive to negative pitch values. The pitch of both outputs is positive, then negative. ADJUST changes both the speed and width of detuning (faster=wider). Stereo In/Stereo Out.
- 3.3 MirrorChorus** **ADJUST: Speed 0–100**  
 An LFO square wave alternates detuning from positive to negative pitch values. The pitch shift of the left output is the opposite of the right. Stereo In/Stereo Out.
- 3.4 LFO Detune** **ADJUST: Detune 0–100**  
 A slow, chorusy detuner. Stereo In/Stereo Out.
- 3.5 SpatialDuck** **ADJUST: cents +/- 0–100**  
 This preset produces a wash of detuned echoes with long reverb decay. The reverb OutWidth is modulated to create spatial movement. The entire effect is ducked by input level. (The reverb and echo decays increase as the level fades.) ADJUST controls the amount of detuning.
- 3.6 Vibrato BPM** **ADJUST: Depth 0–100**  
 The vibrato produced by this effect is tempo controlled so it can be synchronized to the beat. ADJUST sets the vibrato depth. Press **Tap** twice to lock-in the tempo.
- 3.7 Verbato** **ADJUST: Depth 0–100**  
 In this preset, a small stereo chamber reverb is routed through the stereo pitch shifter. The pitch shifter is set to create vibrato. A nice way to thicken up a track without messing with the dry sound.
- 3.8 Stereo VSO** **ADJUST: Rvb Mix 0–100** **Utility Programs (P5 3.8 – P5 3.9)**  
 Use this preset for pitch correcting stereo off-speed playback material. Set the Varispeed parameter (in the first position of the Soft row) to *match* the amount of varispeed used for playback. ADJUST controls the amount of reverb added to the corrected audio.
- 3.9 Mono VSO** **ADJUST: Rvb Mix 0–100**  
 Use this preset for pitch correcting mono off-speed playback material. Set the Varispeed parameter (in the first position of the Soft row) to *match* the amount of varispeed used for playback. ADJUST controls the amount of reverb added to the corrected audio.

**Pitch Correct**  
**(P5 4.0 – P5 4.9)**

With the exception of Demo Correct (4.9), all of the presets are designed to accept a mono vocal track on either (or both) inputs. Note also that, in each preset, the reverb is turned off and all pitch parameters that are not patched can be found in the Soft Row.

- 4.0 VoxFix E2>C6** **ADJUST: On/Off** **0-1**  
 This preset is set up to be used with a MIDI controller with a keyboard, pitch bender and sustain pedal. It provides three different methods to correct pitch problems. You can: use the bender to change the pitch "on the fly," use the sustain pedal to turn correction on and off, or fix a bad note by playing the correct one on the keyboard. Press **Edit** to see the notes displayed. ADJUST is patched to turn correction on and off. All of the pitch parameters that aren't patched are in the Soft Row. Low Note and High Note are tuned to the full vocal range (E2 through C6).
- 4.1 VoxFix Bass** **ADJUST: On/Off** **0-1**  
 The same as VoxFix E2>C6, except that it is tuned to the bass range (E2-A4).
- 4.2 VoxFix Tenor** **ADJUST: On/Off** **0-1**  
 The same as VoxFix E2>C6, except that it is tuned to the tenor range (C3-C5).
- 4.3 VoxFix Alto** **ADJUST: On/Off** **0-1**  
 The same as VoxFix E2>C6, except that it is tuned to the alto range (F3-F5).
- 4.4 VoxFix Sprno** **ADJUST: On/Off** **0-1**  
 The same as VoxFix E2>C6, except that it is tuned to the soprano range (C4-C6).
- 4.5 KnobCentsUp** **ADJUST: Sharp** **0-100**  
 For tracks that are consistently flat. Use ADJUST to tune the track up, and a MIDI keyboard to fix bad notes.
- 4.6 KnobCentsDN** **ADJUST: Flat** **0-100**  
 For tracks that are consistently sharp. Use ADJUST to tune the track down, and a MIDI keyboard to fix bad notes.
- 4.7 Double Effect** **ADJUST: On/Off** **0-1**  
 This preset produces an alternative double track effect. Mix the output of the PCM 81 with the original vocal track. Use ADJUST to randomize the amount of delay and pitch difference between the original and processed tracks.
- 4.8 Knob Freeze** **ADJUST: On/Off** **0-1**  
 This one is a special effect. Turn ADJUST from 0 to 1 to freeze the pitch of the vocal track to whatever was being sung when the knob was turned. Turn it back down to 0 to release the pitch. Mixed with the original, this creates drone-like harmonies. By itself, this effect is a new take on "robot" voices.
- 4.9 Demo Correct** **ADJUST: unused**  
 This preset is for demonstration use. The left input is turned off and all the pitch parameters are in the Soft Row.

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## MIDI Operation

All PCM 81 parameters, programs and registers can be accessed by MIDI. All MIDI applications require the PCM 81 to be connected with one or more MIDI devices with standard MIDI cables via the rear panel MIDI jacks.

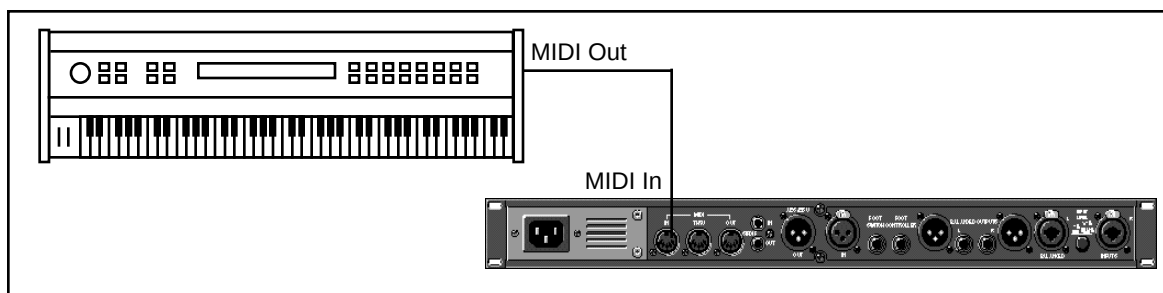
### Selecting a MIDI Channel

MIDI controls, such as Transmit and Receive Channel selection are available in Control mode Row 3 MIDI. All of these controls are described in Chapter 2. Several are repeated here for your convenience.

Before using the PCM 81 with other MIDI devices, all devices must be set to the same MIDI channel. To set the PCM 81 to receive MIDI:

1. Set the controller you will be using (keyboard, sequencer, other PCM 81, etc.) to transmit on any MIDI channel (1-16).
2. On the PCM 81, press **Control**. Use Select and **Up** and **Down** to locate matrix position **3.1 Receive**.
3. Turn ADJUST to select **OFF**, **1-16**, or **OMNI** for receipt of MIDI messages.

### Accessing Programs and Registers

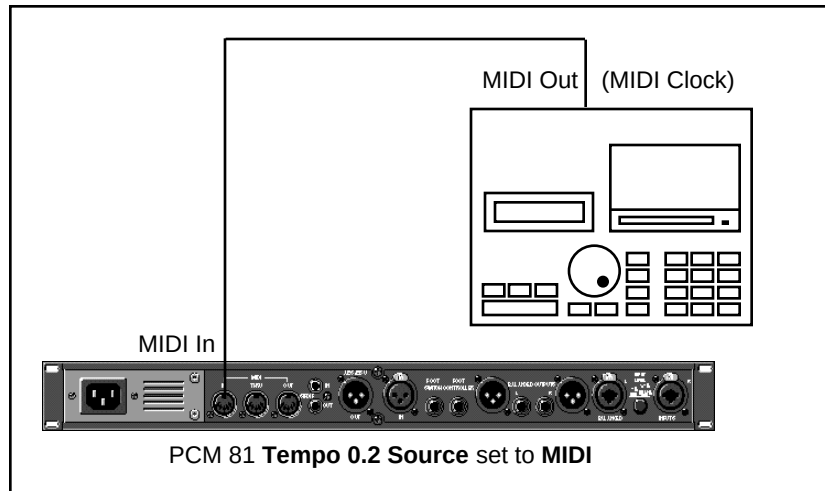


Some extremely useful effects can be created by controlling PCM 81 parameters remotely in real time. Almost all of the controllers found on a MIDI keyboard or MIDI foot controller (pitch benders, mod wheels, sliders, switches, breath controllers, foot pedals and footswitches) can be used to adjust PCM 81 parameters. We refer to this real time remote control capability as Dynamic MIDI®.

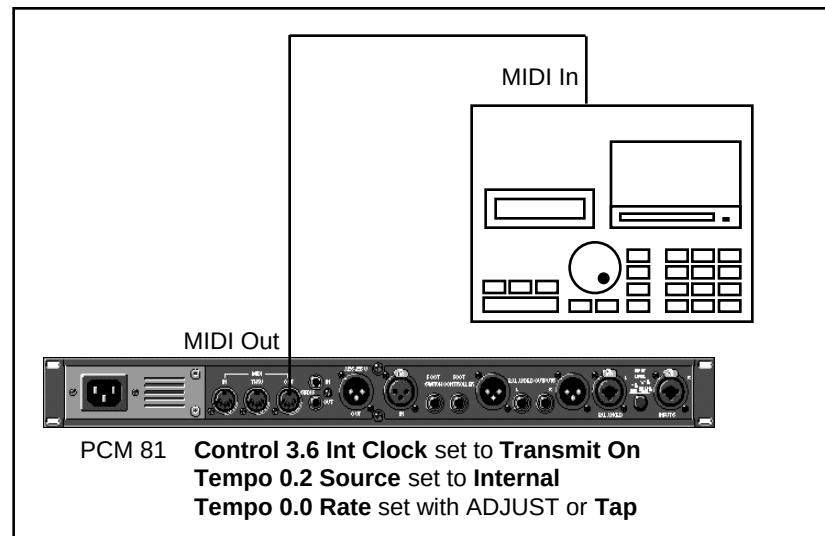
Sending a MIDI Program Change message (0-49) from the controller will load the corresponding PCM 81 register. If any MIDI sources are active as global or general purpose patches, moving the appropriate control on the controller will cause the patched destination parameter to change. (See *Patching*.) If you want to use Dynamic MIDI, but don't want the PCM 81 to load new registers when you change programs on your controller, set your controller so that it doesn't transmit Program Change messages, or set PCM 81 MIDI Program Change to Off at Control mode matrix location 3.3.

## Controlling PCM 81 Tempo Rate with MIDI Clock

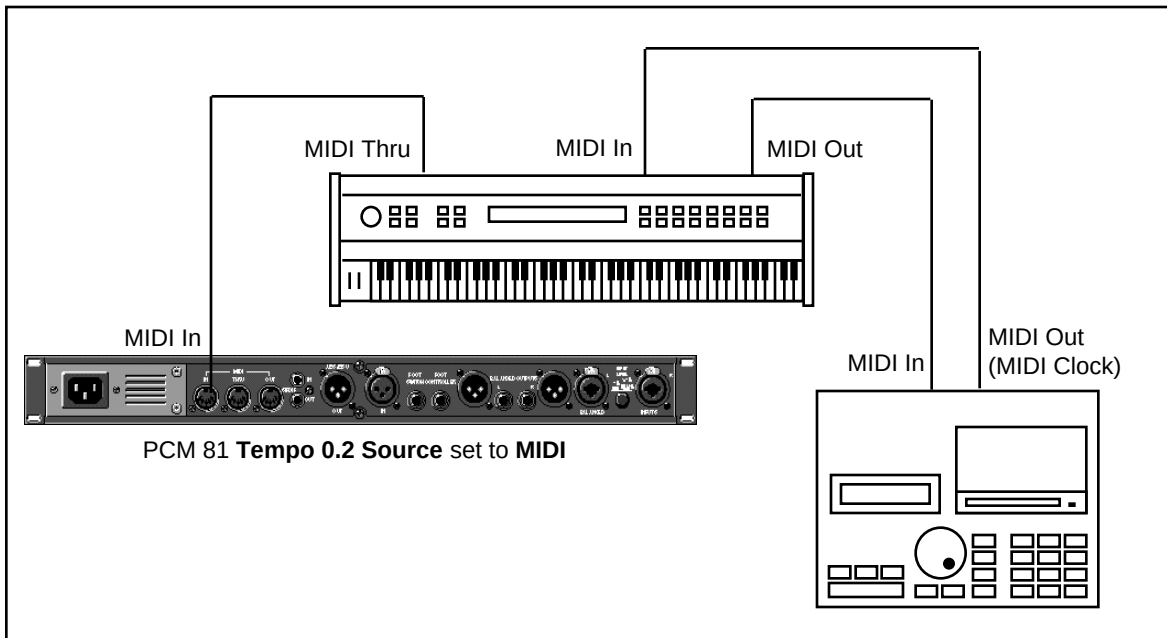
### MIDI Tempo Control



### Using the PCM 81 as a MIDI Clock Source



The configuration below shows the MIDI connections for controlling the PCM 81 simultaneously with MIDI Clocks from a sequencer, and messages from another MIDI controller. Note that the controller is set to "local control off" and the sequencer is set to "echo input".



Two PCM 81s can be slaved together by connecting a cable from the MIDI OUT jack of the master to the MIDI IN jack of the slave. Additional PCM 81s can be slaved to the master by connecting a cable from the MIDI THRU port of one slave unit to the MIDI IN port of the next unit. All of the PCM 81s must be set to the same MIDI channel.

#### Slaving two or more PCM 81s

